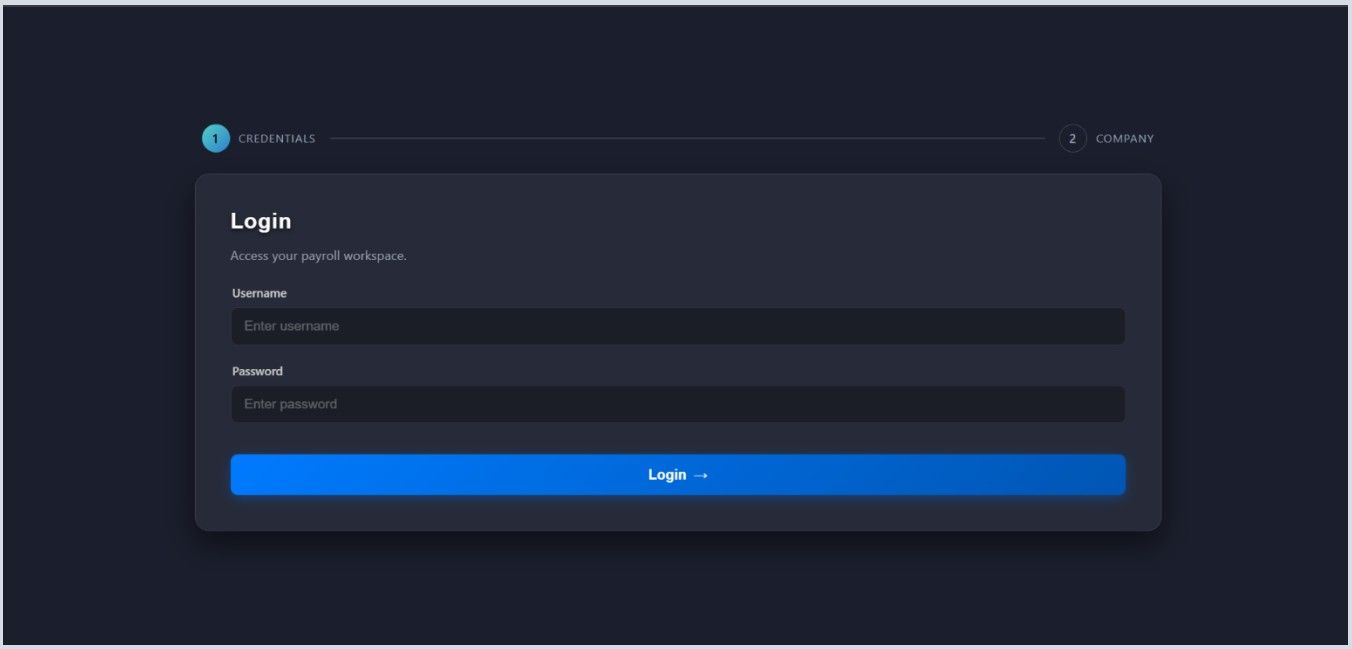
BUSINESS REQUIREMENT SPECIFICATION FOR

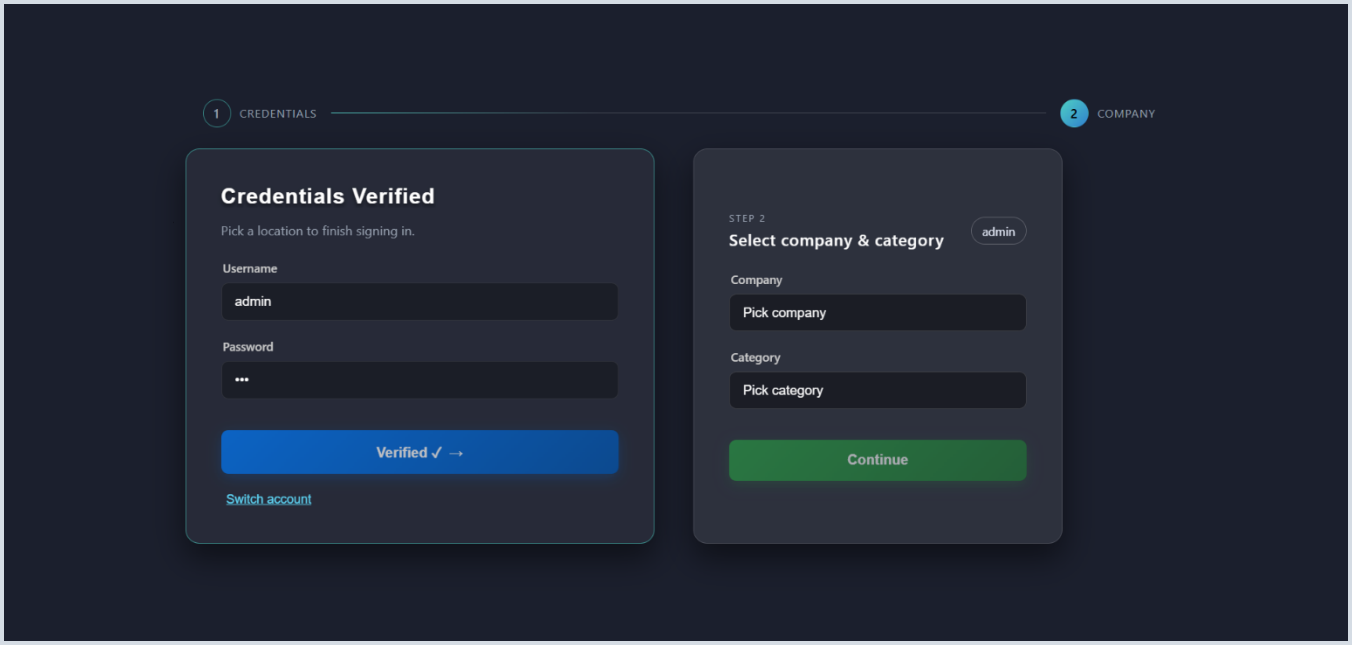
Web-Based Payroll System

VERSION: 1.00

System Overview

The Web-Based Payroll System is a secure, responsive single-page application featuring an intuitive sidebar navigation and modal popups, allowing HR and administrative staff to use the system easily and effectively. The series of activities are categorized and arranged in an orderly, logical flow beginning with Master Files (for Company, Employee, and Bank configurations), followed by daily Transactions (such as loans and salary changes), proceeding to core Processes (for automated month-end calculations including EPF/ETF and dynamic PAYE), and concluding with Reports (digital payslips and bank transfer lists) respectively. This structured layout enables end-users to transition smoothly from legacy workflows and get to the desired screen easily and more quickly.




1 CREDENTIALS
2 COMPANY
Credentials Verified
Pick a location to finish signing in.
Username
admin
Password
•••
Verified ✓ →
Switch account
STEP 2
Select company & category
admin
Company
Pick company
Category
Pick category
Continue

Master Files
Transactions
Processes
Reports
Administration
A
admin
ABC Systems (PVT) Ltd
Managers
Logout

Key System Features & Benefits

Automated Payroll Calculations - Efficiency and accuracy in payroll processing are critical. By automating the monthly calculation cycle, the system eliminates the need for manual, error-prone spreadsheet work. The platform seamlessly processes complex financial transactions, including basic pay adjustments, overtime, attendance allowances, and loan installments, saving the HR team significant time and administrative effort.

Dynamic Tax & Statutory Compliance - Computations for taxes and government statutory deductions can be complex and stressful. This system features automated tax management modules that minimize the burden of employee reporting obligations. All tax data and statutory deductions are calculated automatically and kept securely in the system, making it easy to pull accurate records for compliance without digging through piles of physical pay-slips.

Reliable Backup & Centralized Data Storing large amounts of sensitive payroll data manually or in fragmented files is risky. The application utilizes a centralized database to securely store all employee and financial records. With automated database backups, it is convenient to save records and ensures that, in the event of local hardware failure, all historical payroll data is preserved and fully recoverable.

Comprehensive & Instant Reporting - Generating required month-end documentation is reduced to a few clicks. The system automatically produces formatted digital payslips, customized bank transfer files, and mandatory government EPF/ETF reports. Historical data is securely archived upon month-end finalization, allowing HR to instantly pull past transaction records to quickly resolve any employee disputes.

Web-Based Accessibility & Usability - Unlike standalone desktop applications restricted to specific office computers, this modern web architecture allows authorized staff to securely manage payroll operations from anywhere using a standard web browser. The clean, responsive interface eliminates the steep learning curve of legacy software, making global employee updates and mass data management effortless.

Recommended Hardware & Software Requirements

Because this is a web-based application utilizing a Client-Server architecture, the system requirements are divided between the server environment (where the application and database are hosted) and the client environment (the end-users accessing the system).

Server Environment (Hosting)

Hardware Requirements:

- **Processor (CPU):** Quad-core processor (e.g., Intel Xeon or equivalent) @ 2.5 GHz or higher.
- **Memory (RAM):** 8 GB minimum (16 GB or higher recommended for optimal database performance during heavy month-end processing).
- **Storage:** 100 GB SSD (Solid State Drive) minimum for the operating system, application files, and database storage. SSD is highly recommended to ensure fast read/write speeds for database queries.
- **Network:** High-speed, stable internet connection with a static IP address.

Software Requirements:

- **Operating System:** Linux (e.g., Ubuntu Server 20.04 LTS or newer) or Windows Server (2019 or newer).
- **Web Server / Runtime:** Node.js environment (LTS version).
- **Database Management System:** MySQL (Version 8.0 or higher).
- **Security:** SSL/TLS Certificate for HTTPS encryption.

Client Environment (End-User)

Hardware Requirements:

- **Processor (CPU):** Dual-core processor (Intel Core i3 or equivalent).
- **Memory (RAM):** 4 GB minimum (8 GB recommended for smoother multitasking).
- **Display:** Monitor with a minimum resolution of 1366 x 768 pixels (1920 x 1080 Full HD recommended for optimal viewing of large data grids and reports).
- **Network:** Reliable internet or intranet connection to communicate with the application server.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or modern Linux distributions.
- **Web Browser:** An up-to-date modern web browser is required. Recommended browsers include:
  - Google Chrome (latest version)
  - Mozilla Firefox (latest version)
  - Microsoft Edge (latest version)
  - Apple Safari (latest version)
- **Additional Software:** A standard PDF reader (e.g., Adobe Acrobat or built-in browser PDF viewers) to view and print downloaded reports and payslips.

Non-Functional Requirements

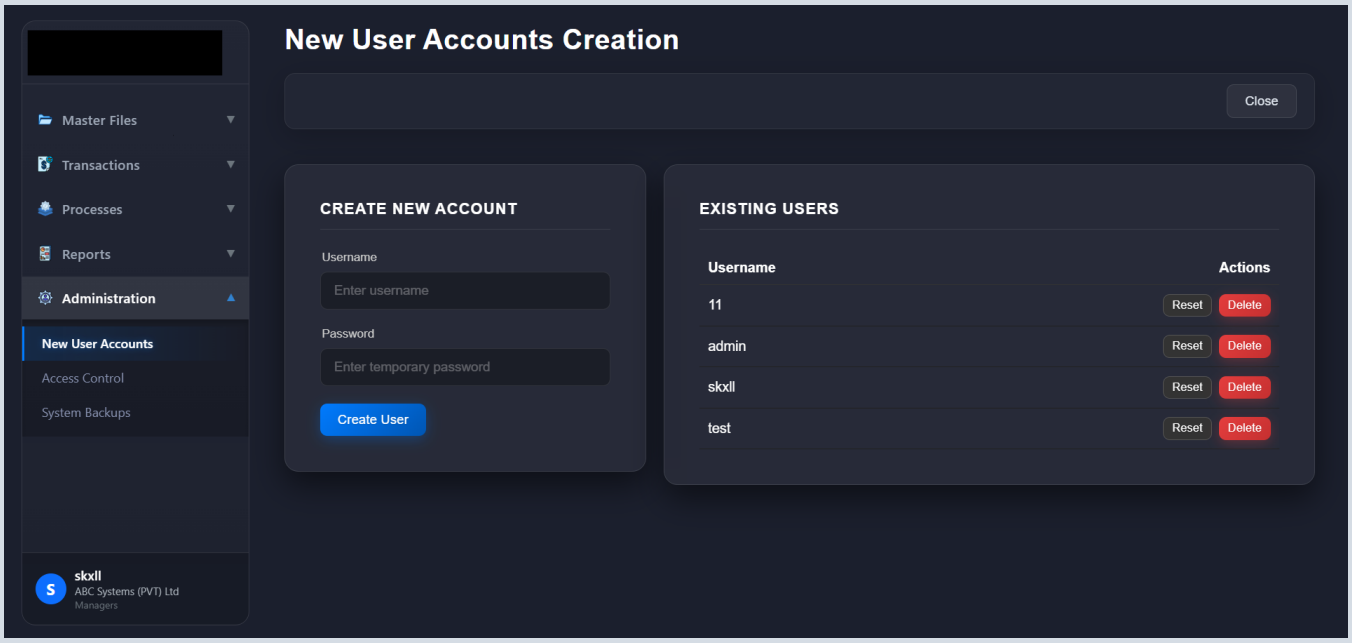
Security

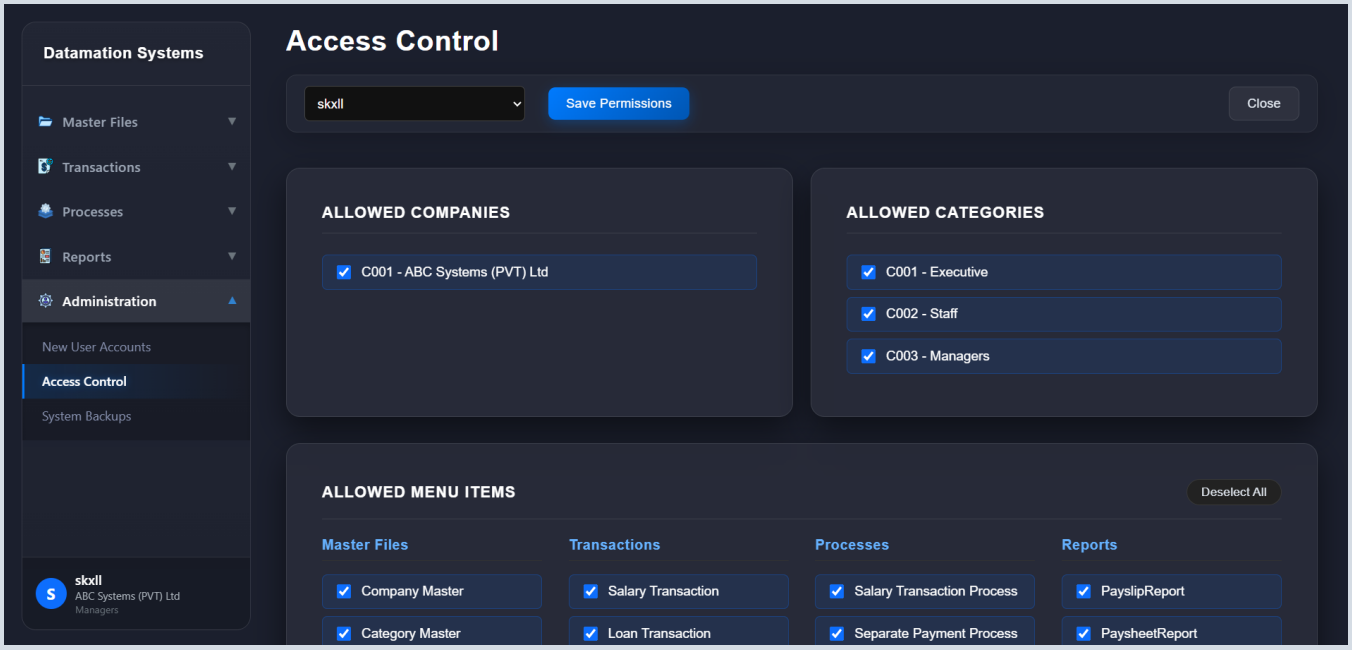
User Creation & Login to the System

- **Account Provisioning:** Only authorized System Administrators will have the capability to create, modify, or deactivate user accounts within the system.
- **Secure Authentication:** The login interface will require users to authenticate using uniquely assigned credentials (e.g., username/email and password). All user passwords must be securely encrypted and hashed before being stored in the database to prevent exposure in the event of a database breach.
- **Session Management:** The application will utilize secure token-based authentication to maintain user sessions. To prevent unauthorized access on unattended terminals, the system will automatically terminate the session and log the user out after a predefined period of inactivity.

Company Access Control

- **Role-Based Access Control (RBAC):** The system will enforce strict RBAC, allowing administrators to configure granular permissions. Users will only have access to the specific menus, screens, and functional operations (e.g., view-only vs. edit/delete capabilities) that are explicitly mapped to their assigned role.
- **Entity-Level Restrictions:** For organizations managing multiple branches, departments, or subsidiary companies, the application will feature dedicated 'Company Access Control'. This ensures that an HR representative or manager assigned to one specific company or department cannot view, edit, or process payroll data for another.
- **Data Segmentation:** Access privileges will be validated at the backend API level, ensuring that even if a user attempts to navigate to a restricted area directly, the system will block the request and protect the underlying data.


New User Accounts Creation
Close
Master Files
Transactions
Processes
Reports
Administration
New User Accounts
Access Control
System Backups
skxll
ABC Systems (PVT) Ltd
Managers
CREATE NEW ACCOUNT
Username
Enter username
Password
Enter temporary password
Create User
EXISTING USERS
Username
Actions
11
admin
skxll
test
Reset
Delete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Access Control
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Master Files
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New User Accounts
Access Control
System Backups
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ABC Systems (PVT) Ltd
Managers
ALLOWED COMPANIES
C001 - ABC Systems (PVT) Ltd
ALLOWED CATEGORIES
C001 - Executive
C002 - Staff
C003 - Managers
ALLOWED MENU ITEMS
Deselect All
Master Files
Transactions
Processes
Reports
Company Master
Category Master
Salary Transaction
Loan Transaction
Salary Transaction Process
Separate Payment Process
PayslipReport
PaysheetReport

Payroll System Operational Cycle

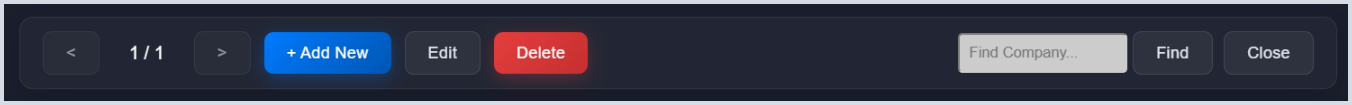
The steps to be followed to run payroll system for a particular month:

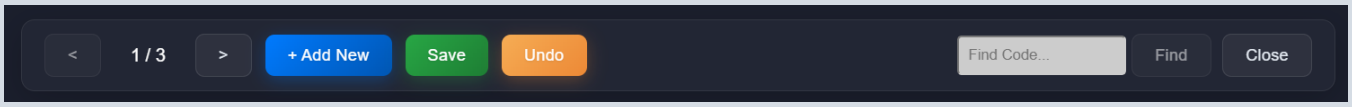
1. Check the system date in the Company master file.
2. Add new employees for this month and edit any existing employees' information in the employee master file.
3. Change any master file information if necessary.
4. Do employee promotion and transfers if available.
5. Enter all monthly data to the "salary transaction".
6. Print "Salary Transaction Listing" report & check for data entry errors.
7. Salary Increments
8. Amend or Add records to the loan transaction
9. Process "Salary Transaction Process"
10. Print all the monthly reports.
11. If user finds any data entry errors, correct it and run the month end again.
12. After finalizing the current month salary data print required reports.
13. Do the banks slip transfer process to send employee salary to bank.
14. EPF ETF e Transfer Process
15. After printing all the reports and to get prepare for the next month, run "Month End Process".

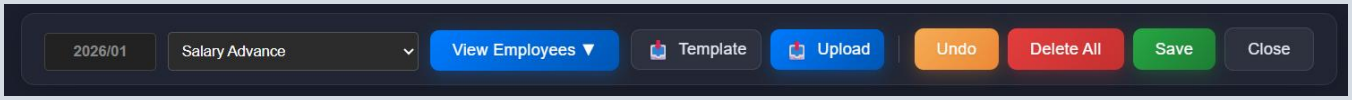
User Interface & Controls

Common Tool Bar Operations to maintain a consistent and intuitive user experience, all data entry screens (such as Master Files and Transaction modules) utilize a standardized action toolbar. These common operations dictate how users interact with the underlying database records.

- **Add / New:** Clears the form to create a new database record.
- **Edit:** Unlocks a selected record to allow modifications.
- **Save / Update:** Validates and commits changes securely to the database.
- **Delete:** Removes a record (includes confirmation prompts to protect financial data).
- **Find:** Searches for specific records within the current data grid.
- **Close:** Exits the active module and returns to the previous screen.
- **View Employees:** Displays a list of employee records relevant to the current process.
- **Template:** Downloads a formatted spreadsheet for bulk data entry.
- **Upload:** Imports a completed template for automated mass processing.
- **Delete All:** Clears all staged records in a temporary batch (requires confirmation).
- **Undo:** Reverts the most recent unsaved action or accidental change.







Functionalities

Module: Company Master File, this interface is used to capture and manage the core structural, statutory, and financial settings for a business entity within the payroll system. The data captured here acts as the foundational master data for processing salaries, generating bank files, and filing statutory reports.

General Details Section This section captures the primary identification and location data for the company.

- Company Code (Mandatory): A unique alphanumeric identifier (e.g., "C001") acting as the primary key reference for the company across the entire application.
- Company Name (Mandatory): The full, legally registered name of the business entity (e.g., "ABC Systems (PVT) Ltd").
- Address: The official registered physical address of the company used for correspondence and official documentation.
- Zone Code: An identifier used to categorize the company into a specific operational, geographical, or administrative zone.
- District Code: A numeric or alphanumeric identifier representing the geographical district where the company is located, often required for localized statutory reporting.

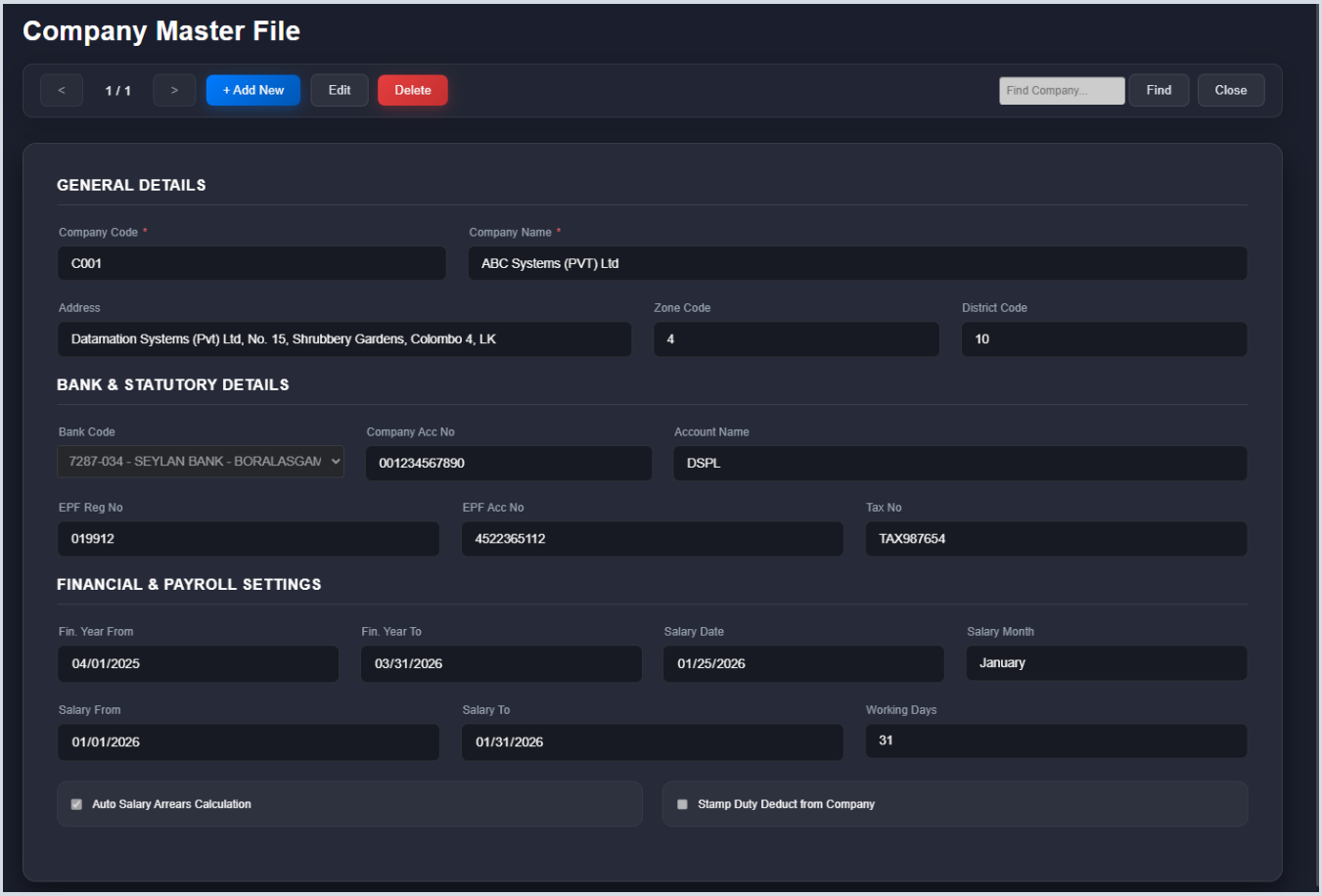
Bank & Statutory Details Section This section stores the financial routing information and government-issued registration numbers necessary for compliance and direct deposits.

- Bank Code: A dropdown or lookup field linking the company to a specific bank and branch (e.g., Seylan Bank) through which payroll fund disbursements will be routed.
- Company Acc No: The official corporate bank account number from which employee salaries and statutory payments will be debited.
- Account Name: The exact legal name registered to the company's bank account.
- EPF Reg No: The Employer Provident Fund registration number assigned to the company by the labor department.
- EPF Acc No: The specific account number associated with the company's EPF contributions and remittance files.
- Tax No: The company's Tax Identification Number (TIN) or VAT number, required for corporate tax and statutory deductions reporting.

Financial & Payroll Settings Section This section defines the active payroll periods, fiscal timelines, and automated calculation rules for the company.

- Fin. Year From: A date field defining the official start date of the company's current financial/fiscal year.

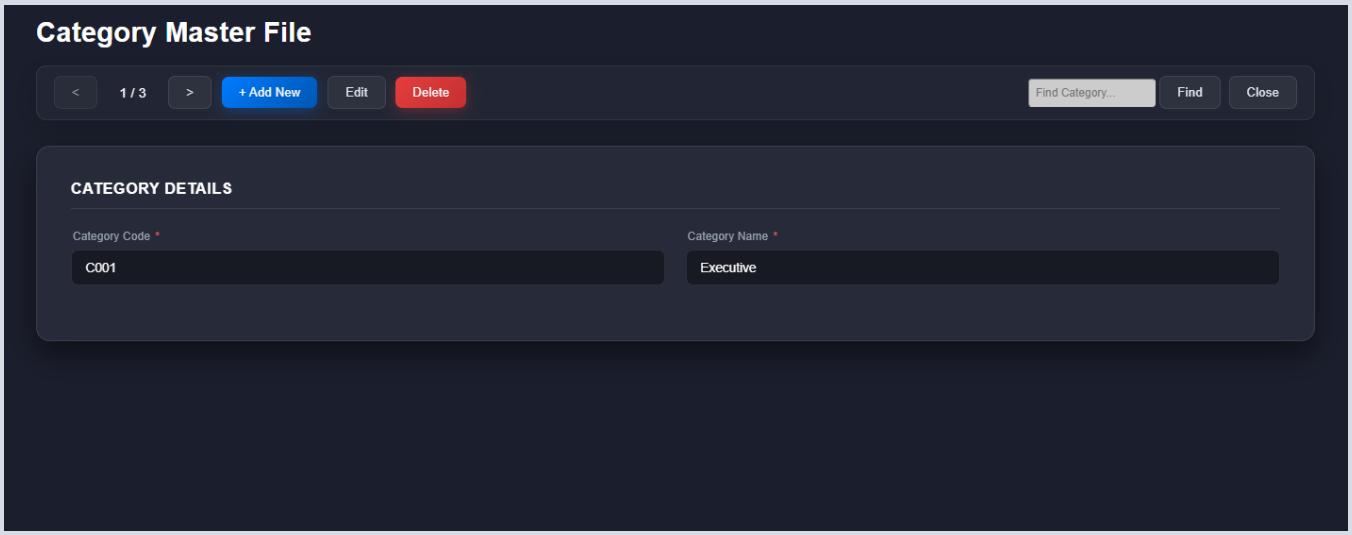
- Fin. Year To: A date field defining the official end date of the company's current financial/fiscal year.
- Salary Date: The standard scheduled date for the physical disbursement of monthly payroll.
- Salary Month: Indicates the current active month that the payroll engine is processing.
- Salary From: The starting date of the current payroll attendance and calculation cycle.
- Salary To: The ending cut-off date of the current payroll attendance and calculation cycle.
- Working Days: The total defined number of billable or payable working days within the current salary period.
- Auto Salary Arrears Calculation (Checkbox): A boolean setting. When enabled, the backend system will automatically calculate and apply any pending back pay, increments, or salary arrears for employees during the monthly payroll execution.
- Stamp Duty Deduct from Company (Checkbox): A boolean setting. When enabled, the statutory stamp duty tax is absorbed as a corporate expense rather than being deducted from the individual employees' net pay.



Module: Category Master File, this interface is used to define and manage employee classifications or groups within the organization (e.g., Executive, Management, Contract, Support Staff). These categories are essential for grouping employees, applying specific payroll rules or benefits, and generating aggregated management reports.

Category Details Section This section captures the core identifying information for the employee classification.

- Category Code (Mandatory): A unique alphanumeric identifier (e.g., "C001") acting as the primary key reference for this specific employee group across the payroll system.
- Category Name (Mandatory): The descriptive, human-readable title of the classification (e.g., "Executive"). This label will appear in employee profiles, dropdown menus, and reporting modules.



Module: Employee Master File, this interface serves as the central repository for individual employee profiles. It aggregates personal identification, organizational placement, compensation parameters, bank routing, and payroll processing rules for each staff member.

Basic Information Section This section captures fundamental personal data and links the employee to the broader organizational structure.

- Company Code (Mandatory): Links the employee to a specific corporate entity defined in the Company Master.
- Category Code (Mandatory): Assigns the employee to a specific classification (e.g., Executive, Contract) defined in the Category Master, determining applicable payroll rules.
- Employee Code (Mandatory): A unique, primary key identifier for the individual employee within the system.
- NIC (Mandatory): The employee's National Identity Card number, essential for legal and statutory identification.
- Initials (Mandatory): The employee's initials or first name(s).
- Surname (Mandatory): The employee's last name or family name.
- EPF No: The employee's specific tracking number for the Employer Provident Fund, used for statutory deductions and reporting.
- Address: The current residential address of the employee.
- Birth Date (Mandatory): The employee's date of birth, used for age validation and retirement calculations.

Employment Details Section This section defines the employee's job role, departmental allocation, and core compensation timeline.

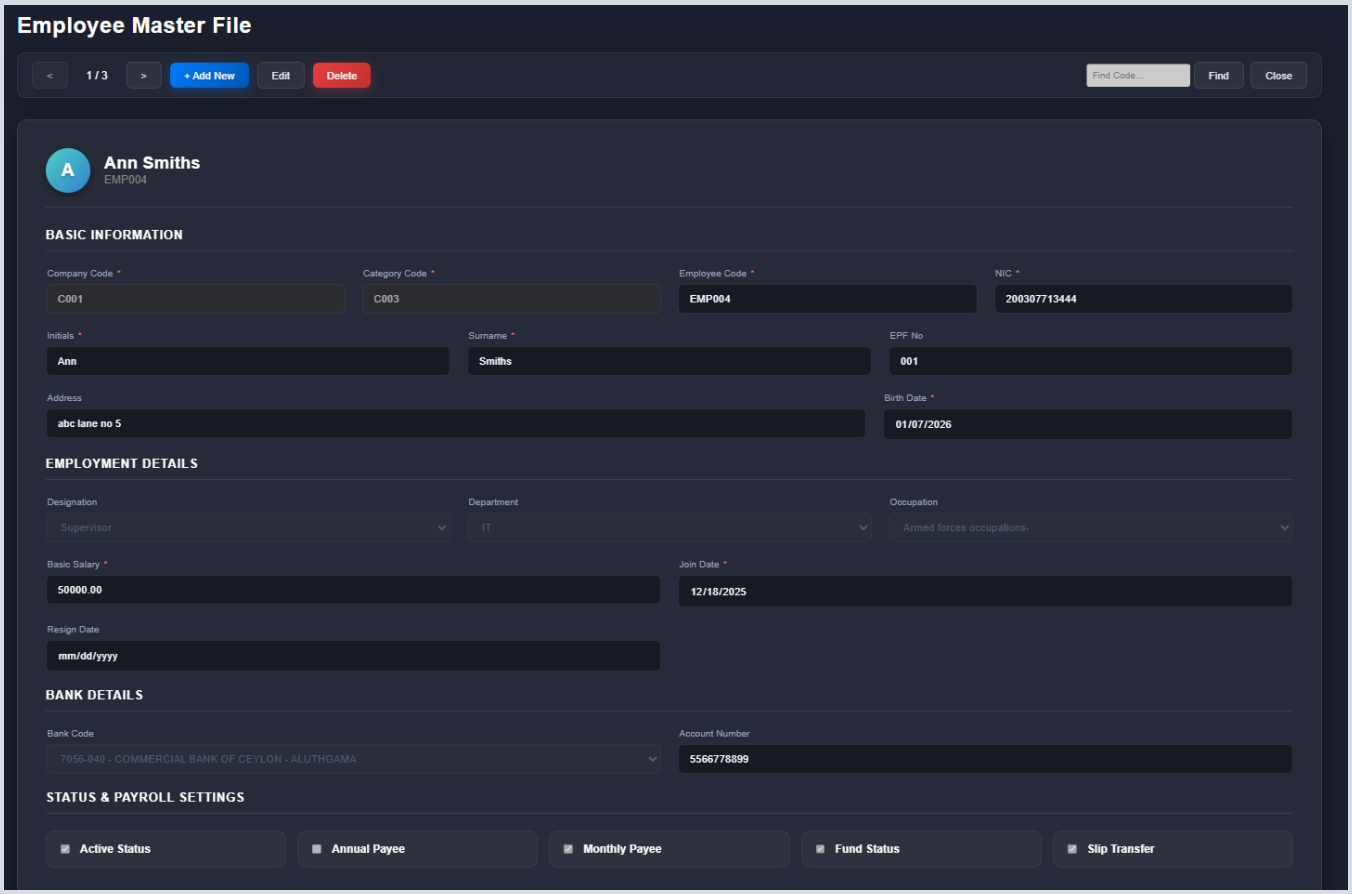
- Designation: A dropdown selecting the employee's official job title (e.g., Supervisor).
- Department: A dropdown assigning the employee to a specific functional group or cost center (e.g., IT).
- Occupation: A dropdown classifying the broader occupational category of the employee, often used for standardized government or statistical reporting.
- Basic Salary (Mandatory): The fixed baseline monetary compensation assigned to the employee before any allowances or deductions are applied.
- Join Date (Mandatory): The official date the employee commenced employment, used to calculate tenure, prorated salaries, and leave entitlements.
- Resign Date: The date the employee's contract was terminated or they resigned. If populated, it acts as a trigger to halt future payroll processing for this individual.

Bank Details Section This section manages the routing information required for automated direct deposits.

- Bank Code: A dropdown linking the employee to a specific bank and branch for salary disbursement.
- Account Number: The employee's personal bank account number where net pay will be credited.

Status & Payroll Settings Section This section contains boolean (checkbox) flags that dictate how the payroll engine handles this specific employee during the processing cycle.

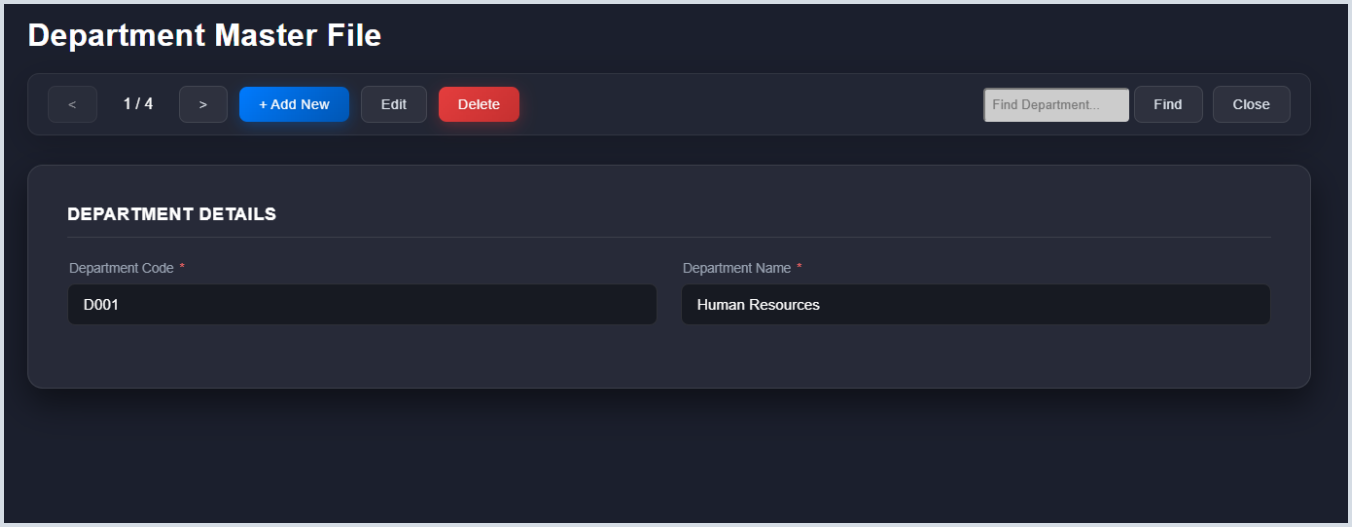
- Active Status: When checked, the employee is considered active and will be included in the current payroll run. If unchecked, they are suspended or inactive and bypassed during processing.
- Annual Payee: Indicates if the employee is subject to annual rather than monthly taxation or salary rules (e.g., specific contractor agreements).
- Monthly Payee: Indicates the employee is on a standard monthly payroll cycle.
- Fund Status: When checked, the system will automatically calculate and deduct statutory fund contributions (like EPF/ETF) for this employee.
- Slip Transfer: When checked, authorizes the automated generation and transfer/emailing of the digital payslip to the employee upon payroll finalization.



Module: Department Master File, this interface is used to define the organizational structure by creating specific departments, divisions, or cost centers within the company. This data allows employees to be grouped logically for management reporting and payroll expense allocation.

Department Details Section

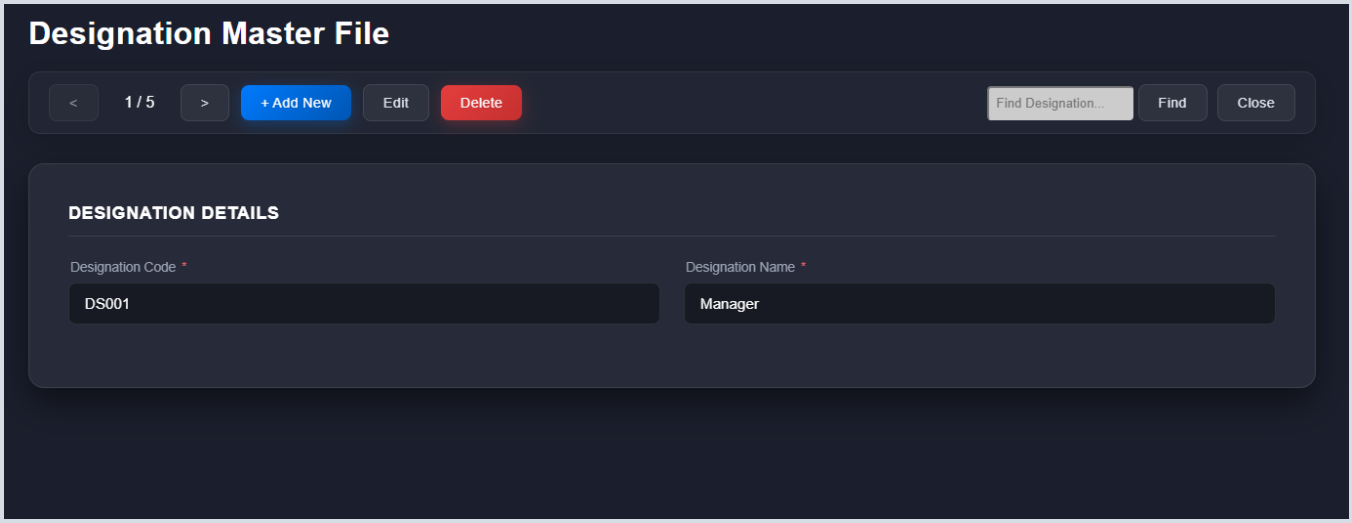
- Department Code (Mandatory): A unique alphanumeric identifier (e.g., "D001") that serves as the primary key for the department in the database.
- Department Name (Mandatory): The standard, human-readable name of the department or cost center (e.g., "Human Resources", "Finance", "IT"). This will populate dropdowns in the Employee Master File and reporting modules.



Module: Designation Master File, this interface manages the standardized list of job titles and organizational roles available within the company.

Designation Details Section

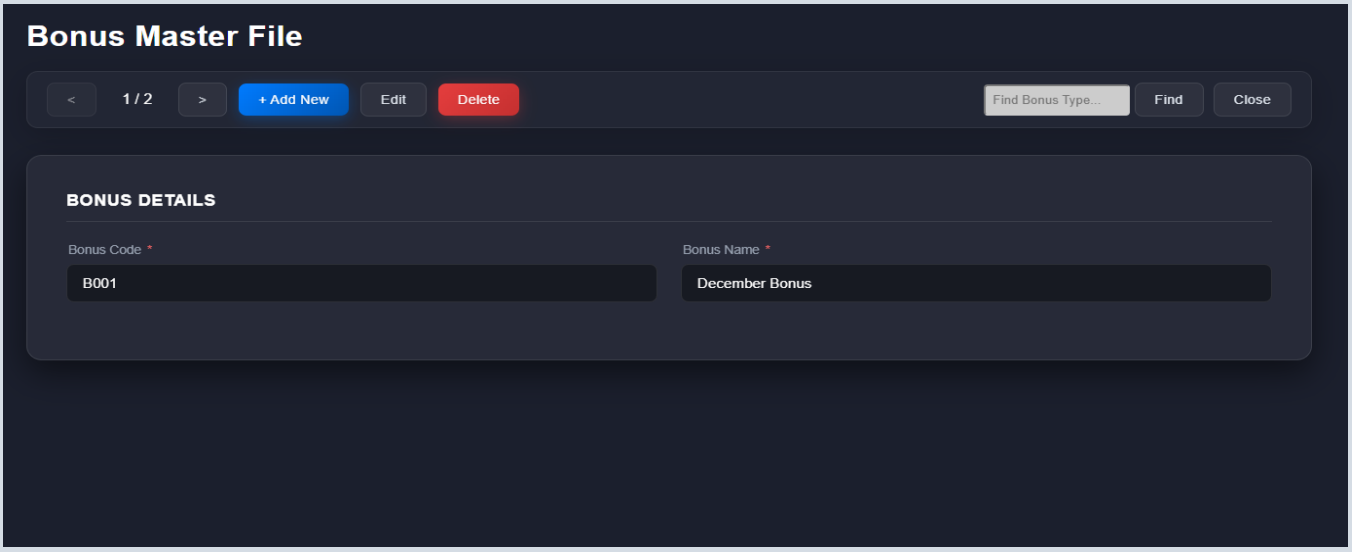
- Designation Code (Mandatory): A unique alphanumeric identifier (e.g., "DS001") acting as the primary key for the specific job role.
- Designation Name (Mandatory): The official job title (e.g., "Manager", "Software Engineer", "Clerk"). This standardized list prevents data entry inconsistencies when assigning roles to employees.



Module: Bonus Master File, this interface defines the various types of non-standard compensation or bonuses that the company can award to employees. Establishing these categories allows the payroll engine to track, calculate, and report on different bonus structures accurately.

Bonus Details Section

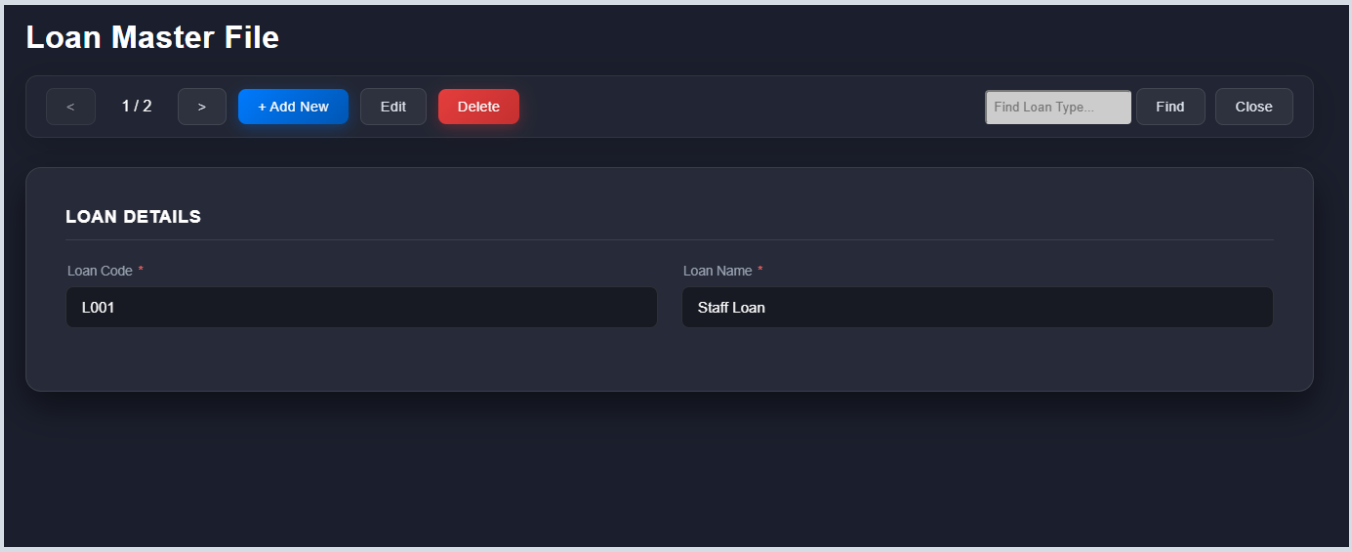
- Bonus Code (Mandatory): A unique alphanumeric identifier (e.g., "B001") used by the backend system to process specific bonus calculations.
- Bonus Name (Mandatory): The descriptive name of the bonus type (e.g., "December Bonus", "Performance Bonus", "Annual Retention Bonus"). This is the label that will appear on the employee's final payslip.



Module: Loan Master File, this interface is used to categorize the different types of financial loans, advances, or credit facilities that the company issues to its employees. This is crucial for managing the distinct recovery rules and deduction schedules associated with different loan types.

Loan Details Section

- Loan Code (Mandatory): A unique alphanumeric identifier (e.g., "L001") representing the specific loan category within the system.
- Loan Name (Mandatory): The descriptive name of the loan type (e.g., "Staff Loan", "Vehicle Loan", "Festival Advance"). This categorizes the deduction when it is processed against the employee's monthly payroll.



Module: Salary Item Master File, this interface is the configuration hub for all financial components that make up an employee's payslip. It allows administrators to define custom earnings, deductions, and allowances, specifying exactly how each item should behave and be calculated during the payroll processing cycle.

Item Details Section This section defines the core attributes and classification of the pay component.

- Company Code (Mandatory): Links the salary item to a specific company entity (e.g., "C001").
- Type (Mandatory): A dropdown classification (e.g., "Earnings", "Deductions") that dictates whether this item will add to or subtract from the employee's gross pay.
- Item Code (Mandatory): A unique identifier for the specific pay element (e.g., "SI105"). This code is heavily used as a reference variable in complex formula building.
- Item Name (Mandatory): The descriptive, human-readable label (e.g., "OT Amount") that will be printed on the employee's final payslip and management reports.
- Pay Date: An optional date field. Used if this specific salary item is scheduled to be disbursed on a date separate from the standard company salary date.

Formula Configuration Section This is the central logic gate for automated payroll processing.

- Formula Configuration (Text Box): A read-only or strictly formatted field that displays the underlying logic structure for the salary item. It is stored as a stringified JSON array, which is optimal for relational database storage and allows a JavaScript-based backend engine to easily parse and evaluate the dynamic expressions during the monthly payroll run.
- Configure Button: Triggers the "Formula Builder" modal, providing a user-friendly interface to construct the JSON logic without requiring manual code entry.

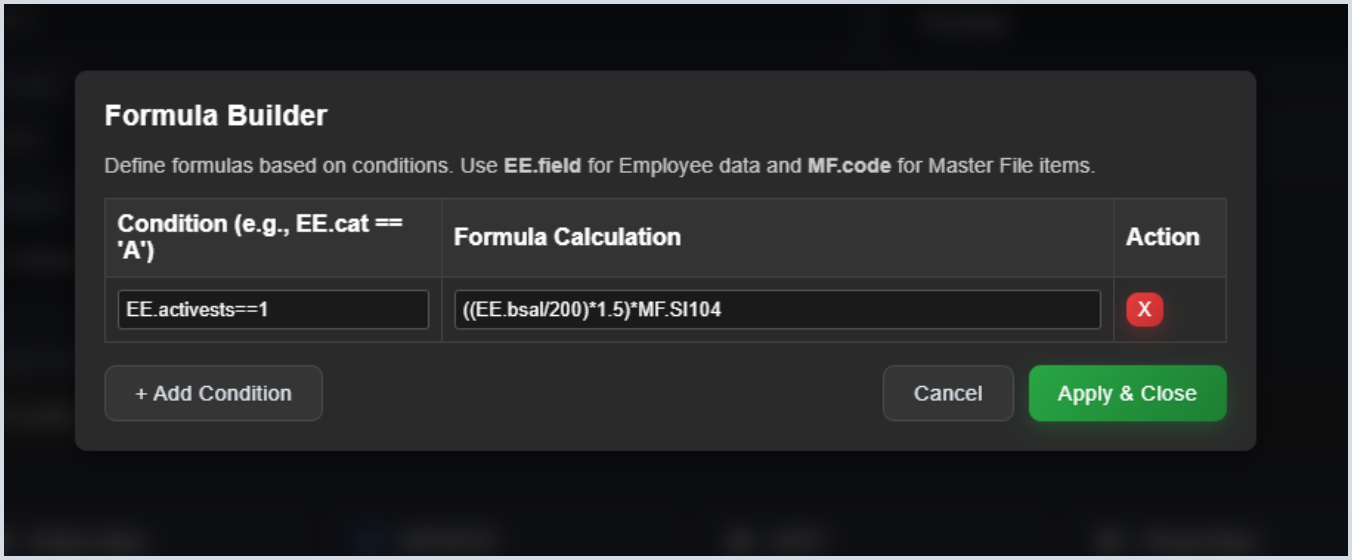
Compliance & Behavior Flags These boolean (checkbox) settings define how the item interacts with statutory bodies and general payroll behavior.

- Entry Item: If checked, this item is not calculated via a formula. Instead, its monetary value is manually entered (or bulk uploaded) each month before payroll processing.
- EPF/ETF: If checked, the monetary value of this item will be included in the gross amount used to calculate statutory Employer Provident Fund and Employees' Trust Fund contributions.
- APIT: If checked, this item is subjected to Advance Personal Income Tax calculations.
- Fixed Item: Indicates that once an amount is set or calculated for an employee, it remains a constant recurring figure in subsequent payroll cycles until manually changed.
- Separate Pay Item: If checked, the system will process and generate a distinct bank transfer file and payslip for this specific item, isolating it from the standard monthly salary.
- Deactivate Item: Acts as a soft-delete mechanism. If checked, the salary item is retained in the database for historical reporting but is ignored by the payroll engine in future calculation runs.

Component: Formula Builder (Modal) A graphical interface accessed via the "Configure" button. It allows administrators to build conditional calculation logic (IF/THEN scenarios) mapping employee data (EE.field) and other master file values (MF.code) to generate dynamic pay amounts.

Formula Definition Grid

- Condition (Input Field): The logical rule that must be met for the formula to execute. It utilizes system variables (e.g., EE.activests==1 checks if the Employee Active Status is true).
- Formula Calculation (Input Field): The mathematical expression executed if the condition is met. It supports standard arithmetic operators and variable fetching.
    - Example Breakdown: ((EE.bsal/200)*1.5)*MF.SI104 fetches the Employee Basic Salary (EE.bsal), divides it by standard working hours (200), multiplies by the overtime rate (1.5), and multiplies the result by the actual OT hours logged for that period (MF.SI104).
- Action (X) Button: Removes the specific condition row from the builder.
- + Add Condition Button: Appends a new row to the grid, allowing administrators to chain multiple scenarios together (e.g., creating tiered commission structures or varying overtime rates based on different conditions).
- Cancel Button: Discards any changes made in the modal and closes it.
- Apply & Close Button: Validates the inputted syntax, serializes the rows into the final JSON array format, populates the parent "Formula Configuration" field, and closes the modal.



Salary Item Master File

< 9 / 9 > + Add New Edit Delete Find Salary Item... Find Close

ITEM DETAILS

Company Code *
C001

Type *
Earnings

Item Code *
SI105

Pay Date
mm/dd/yyyy

Item Name *
OT Amount

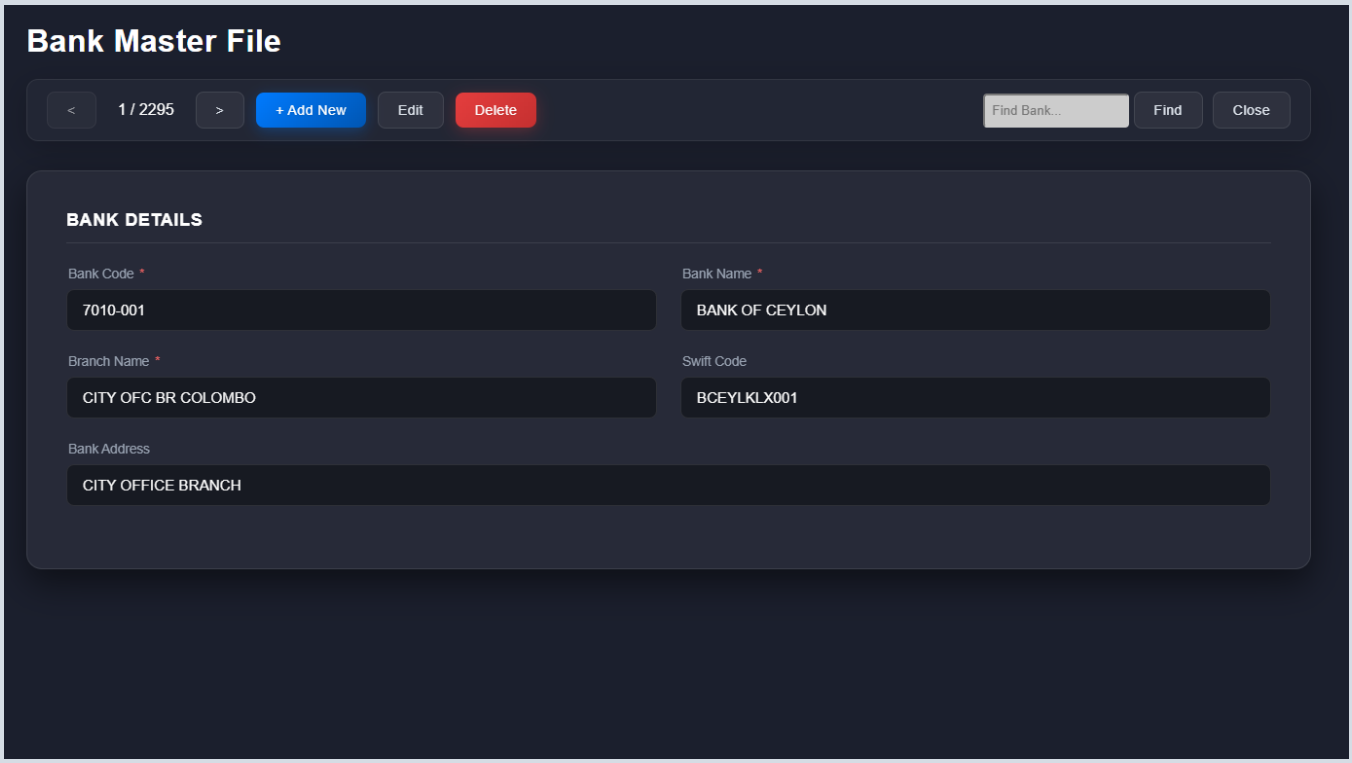
Formula Configuration
[{"condition":"EE.activests==1","formula":"((EE.bsal/200)*1.5)*MF.SI104"}]
Configure

- [ ] Entry Item
- [x] EPF/ETF
- [ ] APIT
- [ ] Fixed Item
- [ ] Separate Pay Item
- [ ] Deactivate Item

Module: Bank Master File, this interface serves as the centralized database for all recognized financial institutions and their respective branches. Maintaining this master list ensures that both company accounts and employee salary disbursements are routed accurately during electronic fund transfers (EFT) and bank file generation.

Bank Details Section This section captures the essential routing and identification data for a specific bank branch.

- Bank Code (Mandatory): A unique identifier used by the system to recognize the specific bank and branch. This is often a composite code (e.g., "7010-001") combining the central bank's assigned institution code with the specific branch routing number. It acts as the primary key for all bank-related lookups.
- Bank Name (Mandatory): The official, legally registered name of the financial institution (e.g., "BANK OF CEYLON").
- Branch Name (Mandatory): The specific, localized name of the bank's branch (e.g., "CITY OFC BR COLOMBO"). This helps distinguish between multiple locations of the same bank.
- Swift Code: The international standard Business Identifier Code (BIC) for the bank. This is particularly crucial for organizations that process international salary transfers or remit payments to foreign employees' accounts.
- Bank Address: The physical location or registered mailing address of the specific bank branch.



Module: APIT Master File, this interface manages the Advance Personal Income Tax (APIT) tables. It allows administrators to define, review, and test the statutory tax brackets and corresponding rates mandated by the government tax authority. The payroll engine references these tables during the calculation phase to determine precise tax deductions for each employee.

Tax Table Selection Controls Unlike standard master files, this screen utilizes toggle buttons to switch between different sets of tax rules based on the payment type.

- Annual Payee Table (Button): Displays the tax brackets and rates applicable to employees whose taxation is calculated on an annualized basis.
- Monthly Payee Table (Button): Displays the standard tax brackets applied to regular monthly salaried employees.
- Bonus Payee Table (Button): Displays the specific, often separate, tax calculation matrix applied strictly to bonus disbursements.

Tax Calculation Testing Utility This is a built-in sandbox feature allowing administrators to verify the system's tax logic without needing to process a live payroll run.

- Enter Amount to Check... (Input Field): A numeric input field where an administrator can enter a hypothetical gross taxable income amount.
- Check Tax (Button): Executes a test calculation using the entered amount against the currently active tax table (Annual, Monthly, or Bonus). The system will return the calculated tax value, allowing admins to instantly verify that the tax configurations are accurate and compliant.

Tax Bracket Data Grid

This table defines the actual mathematical boundaries for the tax calculations.

- RANGE FROM: The minimum monetary threshold (starting value) for a specific tax bracket.
- RANGE TO: The maximum monetary threshold (upper limit) for that specific tax bracket.
- TAX RATE (%): The specific percentage or defined tax value applied to the portion of the employee's income that falls strictly within this defined "From" and "To" range.

APIT Master File

Annual Payee Table | Monthly Payee Table | Bonus Payee Table

Enter Amount to Check... | Check Tax | Close

RANGE FROM	RANGE TO	TAX RATE (%)
0	1,800,000.99	0
1,800,001	1,800,200.99	12
1,800,201	1,800,400.99	24
1,800,401	1,800,600.99	36
1,800,601	1,800,800.99	48
1,800,801	1,801,000.99	60
1,801,001	1,801,200.99	72
1,801,201	1,801,400.99	84
1,801,401	1,801,600.99	96
1,801,601	1,801,800.99	108
1,801,801	1,802,000.99	120
1,802,001	1,802,200.99	132
1,802,201	1,802,400.99	144
1,802,401	1,802,600.99	156
1,802,601	1,802,800.99	168
1,802,801	1,803,000.99	180

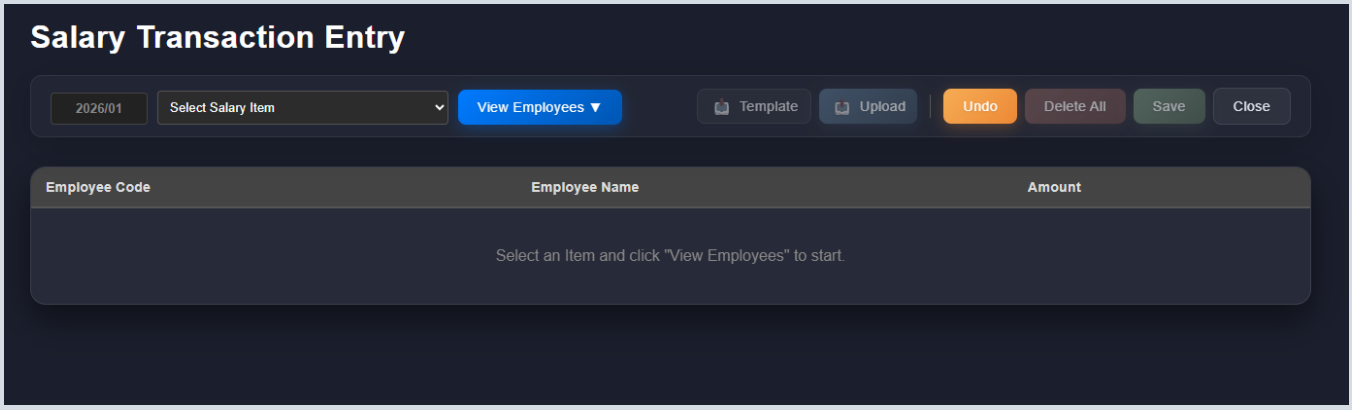
Module: Salary Transaction Entry, this interface is the central operational hub for managing and adjusting values for any salary component within a specific payroll month. It allows administrators to manually enter, bulk-upload, or override amounts for all salary items defined in the Salary Item Master (including entry items, fixed items, and calculated variables) before the final payroll calculation engine is executed.

Selection & Batch Controls This top section dictates which payroll period and specific salary component are currently being processed.

- Payroll Period Indicator (e.g., 2026/01): A read-only display showing the active financial year and salary month for which these transactions will be applied.
- Transaction Type (Dropdown): Allows the user to select the specific salary item they are targeting. This dropdown is populated by active items from the Salary Item Master File, regardless of whether they are fixed, variable, or entry items.
- View Employees (Button): Executes a query to populate the data grid below with active, eligible employees. Crucially, if values have already been added or calculated for specific employees for the selected item, those employees will load into the grid with their existing amounts pre-filled, allowing the user to review or manipulate the data.

Data Entry Grid This section provides a tabular view for rapid data entry and modification once an item is selected.

- Employee Code: A read-only display of the employee's unique system identifier.
- Employee Name: A read-only display of the employee's full name.
- Amount (Input Field): A numeric input field. It displays the pre-existing value if data is already logged for this employee. Administrators can manually type here to input a new value or adjust/override the existing data for the current payroll cycle.



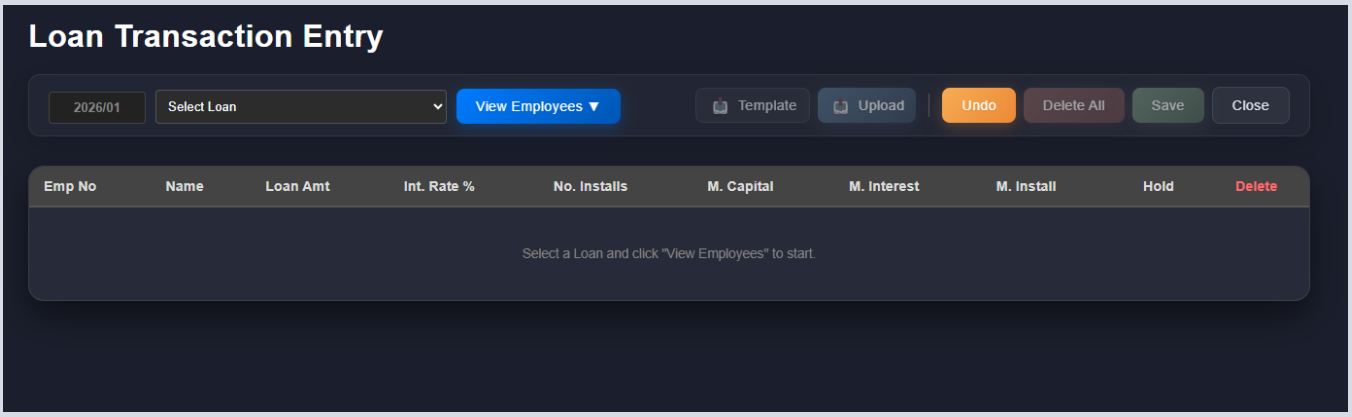
Module: Loan Transaction Entry, this interface is used to issue new loans to employees and configure their specific recovery schedules. It dictates how the payroll engine will automatically deduct monthly installments from the employee's gross pay.

Selection Controls

- Payroll Period Indicator: The active financial year and month the loan or deduction commences.
- Select Loan (Dropdown): Allows the administrator to choose the specific loan category (e.g., Staff Loan, Vehicle Loan) defined in the Loan Master File.
- View Employees (Button): Populates the grid with employees. If an employee already has an active loan of this type, their existing schedule will load for review or modification.

Data Entry Grid

- Emp No & Name: Read-only displays of the employee's identifiers.
- Loan Amt: The total principal principal amount issued to the employee.
- Int. Rate %: The specific interest rate applied to this individual's loan.
- No. Installs: The total number of monthly periods over which the loan will be recovered.
- M. Capital: The calculated or manual monthly deduction applied to the principal balance.
- M. Interest: The calculated or manual monthly deduction for interest.
- M. Install: The total monthly deduction amount (Capital + Interest) that will be subtracted from the employee's salary.
- Hold (Checkbox): A critical operational toggle. If checked, the system will temporarily pause the loan deduction for the current payroll processing month without deleting the overall loan record.
- Delete (Action): Removes the specific loan entry for that employee from the system.



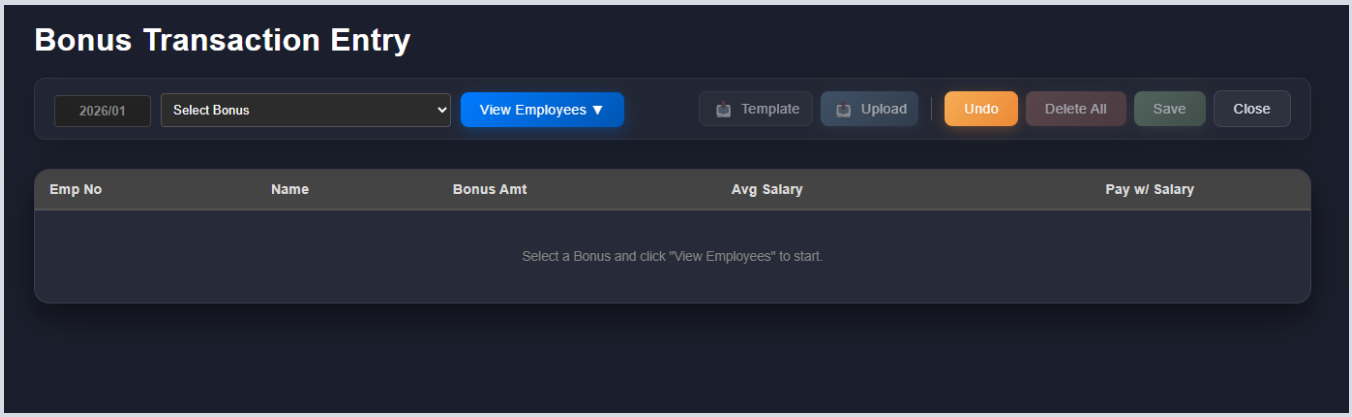
Module: Bonus Transaction Entry, this interface manages the allocation of non-standard compensation, allowing administrators to input or override bonus amounts before the final payroll run.

Selection Controls

- Payroll Period Indicator: The active financial year and month for the bonus payout.
- Select Bonus (Dropdown): Allows selection of the specific bonus type (e.g., Annual Bonus, Performance Bonus) configured in the Bonus Master File.

Data Entry Grid

- Emp No & Name: Read-only employee identifiers.
- Bonus Amt: The precise monetary value of the bonus awarded to the employee.
- Avg Salary: A read-only reference field displaying the employee's calculated average salary, as certain corporate bonuses are calculated as multiples or percentages of this figure.
- Pay w/ Salary (Checkbox): A routing flag. If checked, the bonus amount is merged into the standard monthly paycheck and bank file. If unchecked, the system flags this amount to be disbursed via a separate, dedicated bank transfer.



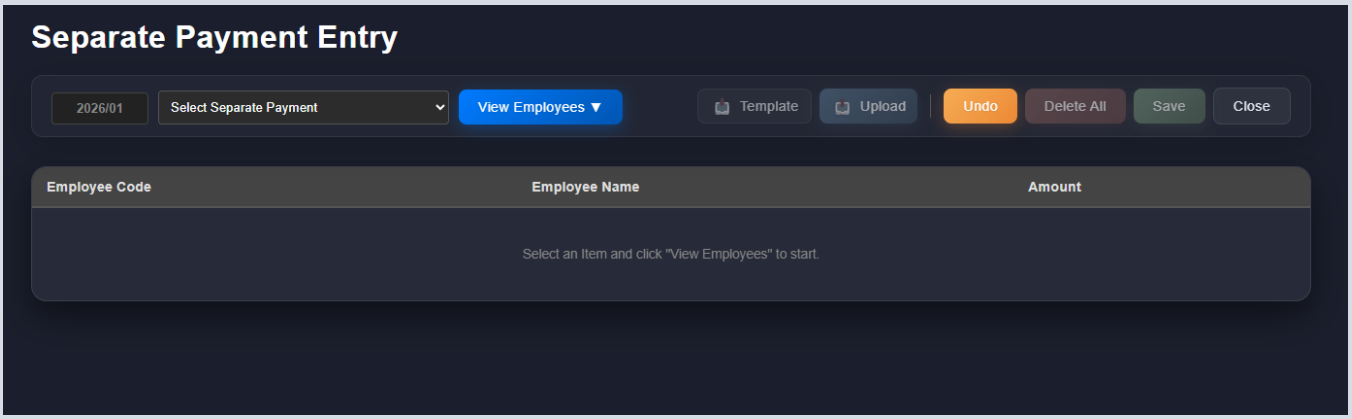
Module: Separate Payment Entry, this simplified interface is used to process ad-hoc disbursements that must remain strictly isolated from the standard monthly salary calculations (e.g., specific expense reimbursements, severance packages, or specialized allowances).

Selection Controls

- Payroll Period Indicator: The active financial period.
- Select Separate Payment (Dropdown): Allows the user to select an item flagged specifically as a "Separate Pay Item" in the Salary Item Master File.

Data Entry Grid

- Employee Code & Employee Name: Read-only employee identifiers.
- Amount: The monetary value of the separate payment to be processed.



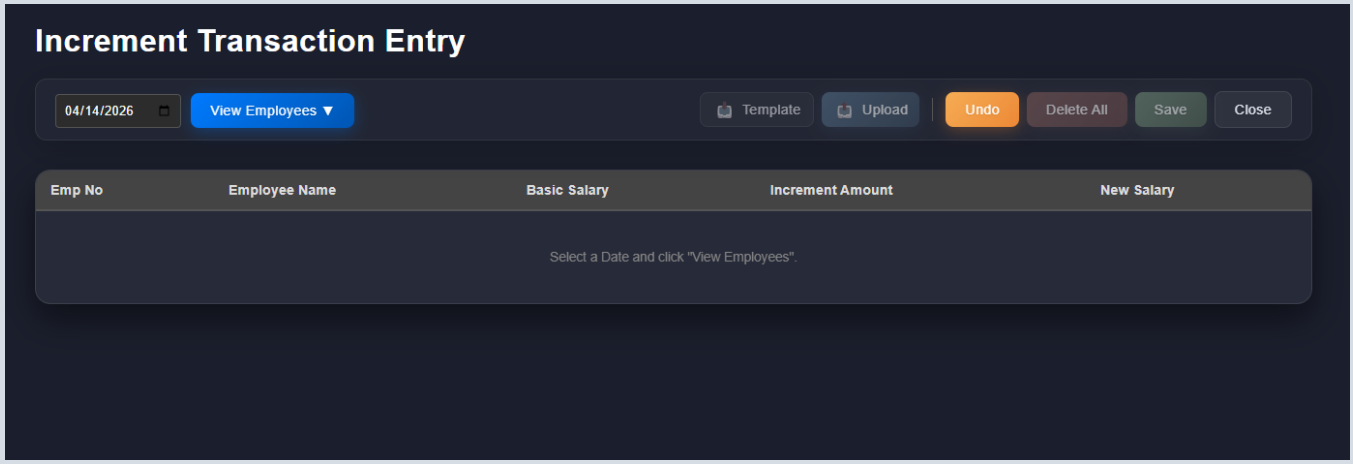
Module: Increment Transaction Entry, this module processes organizational salary raises. Unlike other transaction screens that affect a single month's pay, this module permanently updates the employee's foundational Basic Salary in the Employee Master File.

Selection Controls

- Effective Date (Date Picker): Replaces the standard period indicator. This field defines the exact date the new salary rate becomes legally active (e.g., 04/14/2026). The payroll engine uses this date to calculate prorated arrears if the increment is backdated.

Data Entry Grid

- Emp No & Employee Name: Read-only employee identifiers.
- Basic Salary: A read-only display of the employee's current baseline compensation before the increment is applied.
- Increment Amount: The specific monetary value being added to the employee's current salary.
- New Salary: A calculated, read-only field displaying the final, updated compensation figure (Basic Salary + Increment Amount) that will be written to the employee's master profile upon saving.



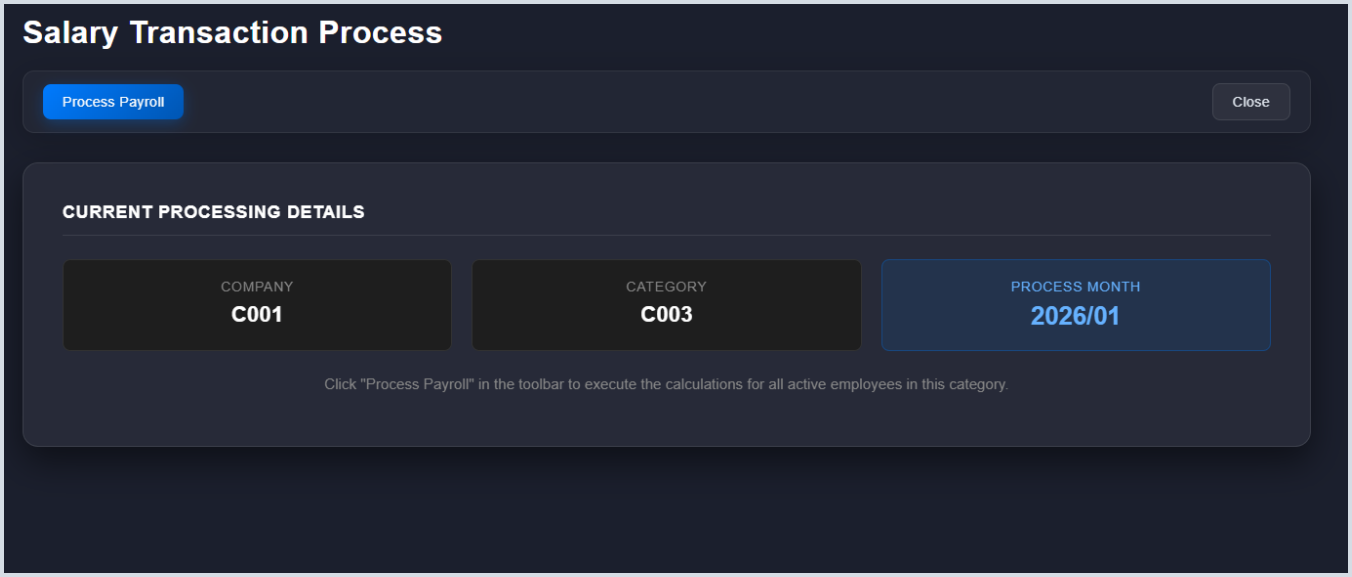
Module: Salary Transaction Process, this interface serves as the central execution trigger for the main monthly payroll engine. It initiates the backend calculation sequence that processes all active employees, applies their relevant salary item formulas, factors in manual transaction overrides, and generates the final payslip data for the selected period.

Action Bar

- Process Payroll (Button): The primary execution command. Clicking this triggers the system to run the complex calculation algorithms for the defined parameters.
- Close (Button): Exits the module and returns the user to the previous screen.

Current Processing Details Section This section displays read-only contextual data, confirming the exact parameters the calculation engine will use before the user commits to processing.

- COMPANY (Read-only): Displays the specific Company Code (e.g., "C001") that is currently queued for processing.
- CATEGORY (Read-only): Displays the specific Employee Category Code (e.g., "C003") targeted for this batch run. This allows administrators to process payroll in segmented batches (e.g., running Executives separately from Contract staff).
- PROCESS MONTH (Read-only): Displays the active financial year and month (e.g., "2026/01") for which the salary calculations are being executed.
- Helper Text: Provides a clear system instruction to the user: "Click 'Process Payroll' in the toolbar to execute the calculations for all active employees in this category."



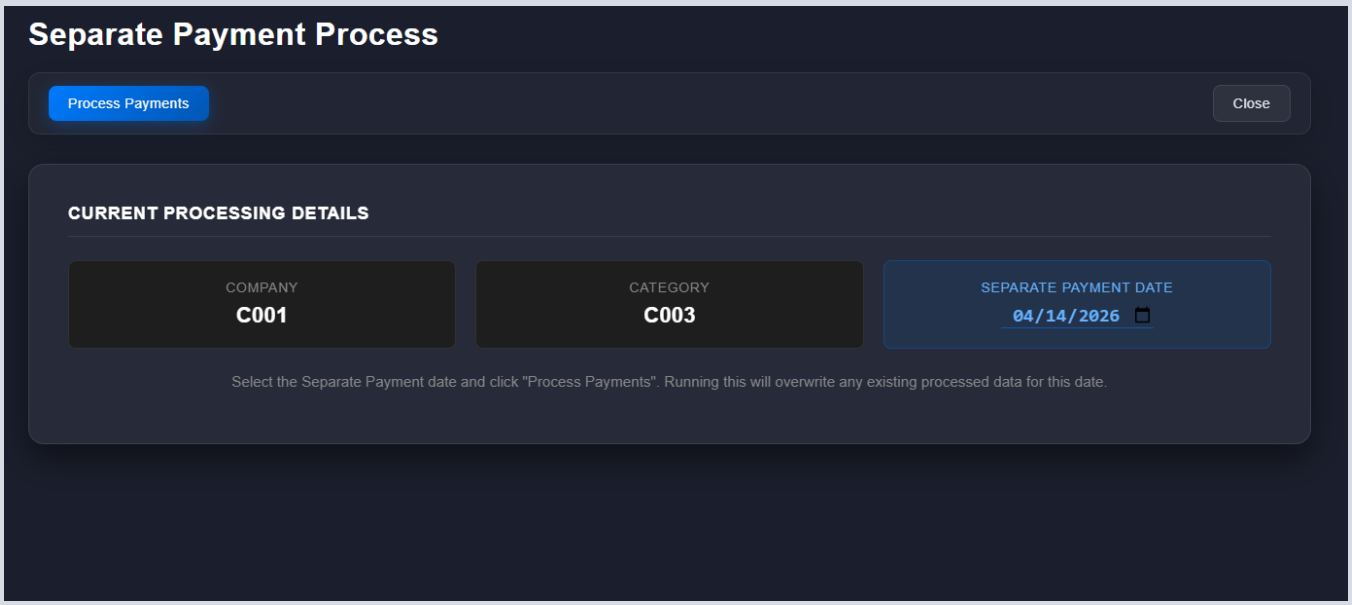
Module: Separate Payment Process, this interface triggers the calculation engine strictly for ad-hoc, isolated disbursements that were logged in the "Separate Payment Entry" module. It generates the final payable amounts for these specific items, ensuring they remain independent of the standard monthly salary calculations and bank files.

Action Bar

- Process Payments (Button): The primary execution command to run the backend calculations specifically for the separated pay items.
- Close (Button): Exits the module.

Current Processing Details Section This section confirms the scope of the ad-hoc payment run.

- COMPANY (Read-only): Displays the Company Code for the batch.
- CATEGORY (Read-only): Displays the Employee Category targeted for this separate payment run.
- SEPARATE PAYMENT DATE (Date Picker): Unlike the standard payroll which runs on a monthly block, this is a selectable date field. The administrator must pick the exact date these separate payments are scheduled to be disbursed. The engine will only process separate payment entries tied to this specific date.
- Helper Text: Provides critical system behavior information: "Select the Separate Payment date and click 'Process Payments'. Running this will overwrite any existing processed data for this date." This warns the user that rerunning the process for the same date will recalculate and replace any previously generated figures.



Module: Slip Transfer Process, this interface is the final step in the disbursement cycle. Once payroll calculations are processed and finalized, this module is used to extract that calculated data and generate the electronic bank transfer files (such as EFT or ACH text files). These files are then uploaded directly to the company's corporate banking portal to execute the physical transfer of funds into the employees' personal bank accounts.

Action Bar

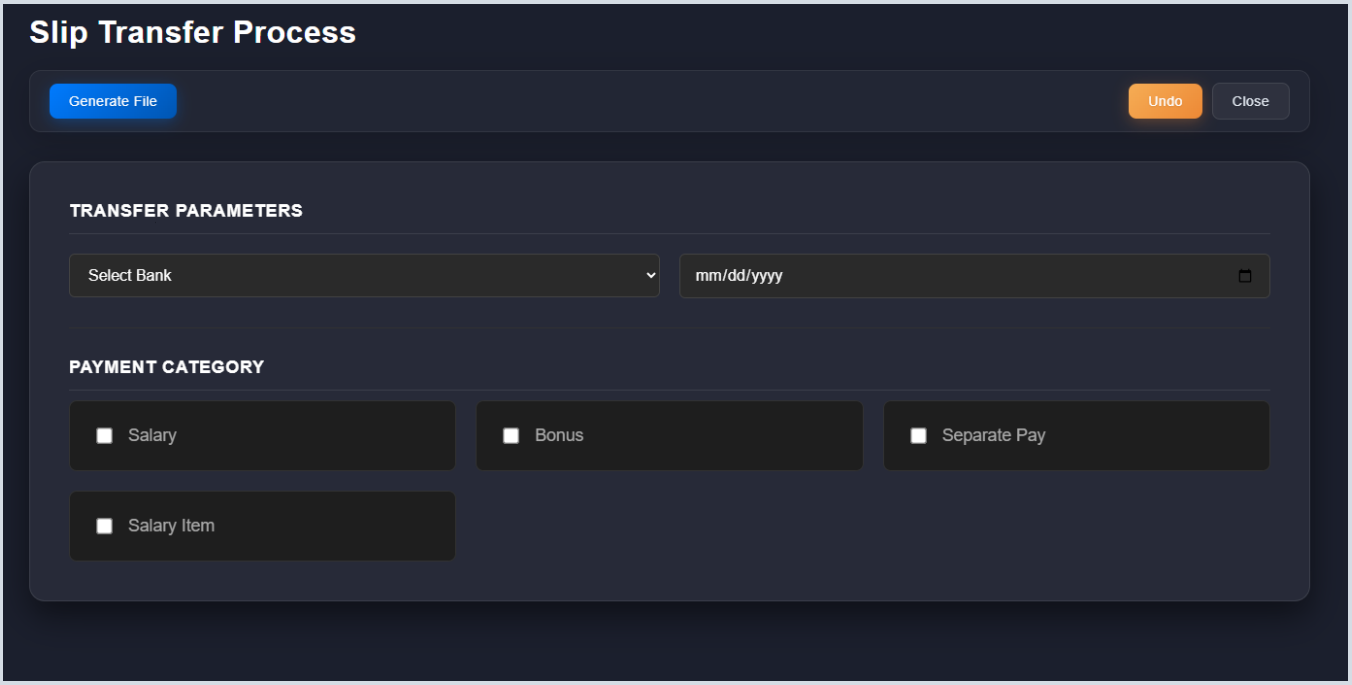
- Generate File (Button): The primary execution command. Clicking this compiles the selected payroll data and downloads the formatted bank transfer file.
- Undo (Button): Reverts or cancels the most recent file generation batch if an error was made before the file was officially submitted to the bank.
- Close (Button): Exits the module and returns the user to the previous dashboard.

Transfer Parameters Section This section defines the routing rules and timeline for the bank file.

- Select Bank (Dropdown): Allows the administrator to choose the specific corporate bank account from which the funds will be debited. This selection also dictates the specific file format and data structure generated by the system, ensuring it complies with that exact bank's upload requirements.
- Transfer Date [mm/dd/yyyy] (Date Picker): Specifies the "Effective Date" or execution date. This tells the bank exactly when the funds should be cleared and deposited into the employees' accounts.

Payment Category Section This section acts as a data filter, allowing administrators to dictate exactly which types of processed payments should be included in the generated bank file.

- Salary (Checkbox): If checked, the file will include the standard, finalized monthly net pay for all eligible employees.
- Bonus (Checkbox): If checked, includes any processed bonus amounts. This is particularly useful if the company opts to disburse bonuses in a separate bank run rather than merging them with the monthly salary.
- Separate Pay (Checkbox): If checked, includes ad-hoc disbursements that were calculated via the "Separate Payment Process" module.
- Salary Item (Checkbox): If checked, allows the extraction of specific, isolated salary components that are configured in the master files to be paid out independently.



Module: EPF Process, this interface handles the statutory compliance aspect of the payroll cycle. After salaries are finalized, this module extracts the combined Employee Deductions and Employer Contributions for the Employer Provident Fund (EPF) and generates the officially formatted electronic return file. This file is then uploaded to the relevant government or central bank portal to remit the statutory payments.

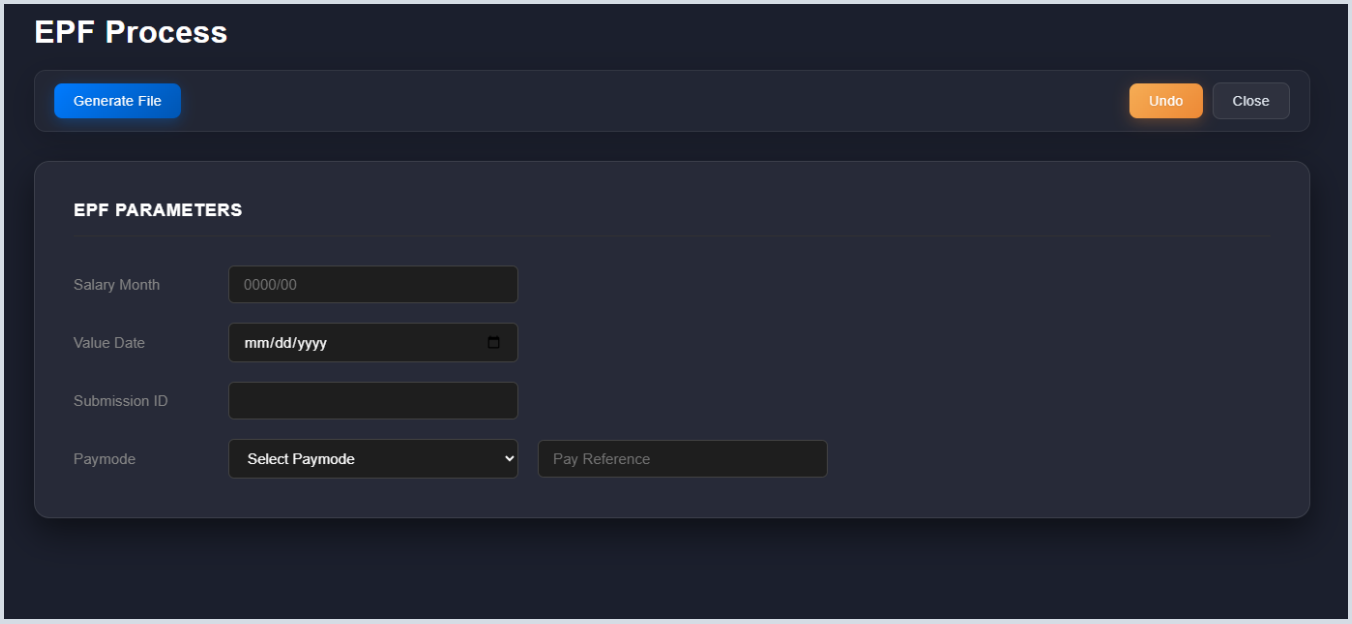
Action Bar

- Generate File (Button): The primary execution command. Clicking this compiles the EPF data based on the defined parameters and downloads the specific text or CSV file required by the statutory authority.
- Undo (Button): Reverts or cancels the most recently generated batch record, typically used if an error is discovered before the file is officially uploaded to the government portal.
- Close (Button): Exits the module and returns the user to the previous dashboard.

EPF Parameters Section This section defines the financial tracking and routing details that will be embedded into the header of the generated statutory file.

- Salary Month (Input Field): Specifies the exact financial year and month (e.g., "2026/01") for which these contributions were deducted. This ensures the system only pulls EPF data from that specific, finalized payroll run.

- Value Date (Date Picker): The effective date of the financial transaction. This tells the statutory body's system on exactly what date the funds will be cleared and debited from the corporate bank account.
- Submission ID (Input Field): A unique alphanumeric tracking number assigned to this specific batch or file upload. This is often a mandatory field required by the receiving government portal to track and audit the return.
- Paymode (Dropdown): Defines the method by which the company is transferring the physical funds to the EPF board (e.g., Direct Debit, Cheque, Online Transfer, Cash).
- Pay Reference (Input Field): A supplementary tracking field tied to the chosen Paymode. If the payment is made via cheque, the cheque number is entered here; if by online transfer, the bank's transaction reference code is used. This links the electronic return file to the actual physical movement of funds.



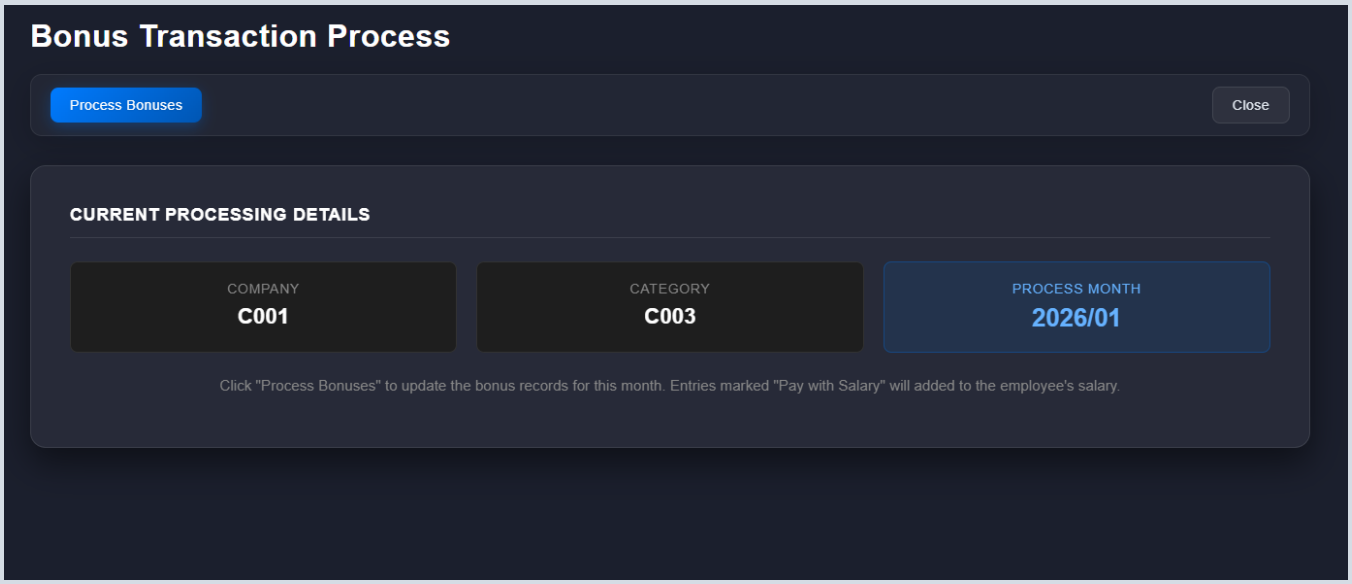
Module: Bonus Transaction Process, this interface serves as the execution trigger specifically for finalizing bonus disbursements. It processes the raw data previously inputted in the "Bonus Transaction Entry" module, applies any relevant tax calculations (referencing the APIT Bonus tables), and routes the final amounts based on how they were configured to be paid.

Action Bar

- Process Bonuses (Button): The primary execution command. Clicking this initiates the backend script that updates and locks the bonus records for the selected period.
- Close (Button): Exits the module and returns the user to the previous screen.

Current Processing Details Section This section displays read-only contextual data, confirming the exact scope of the calculation engine before the user commits to processing the batch.

- COMPANY (Read-only): Displays the specific Company Code (e.g., "C001") currently queued for bonus processing.
- CATEGORY (Read-only): Displays the specific Employee Category Code (e.g., "C003") targeted for this batch run.
- PROCESS MONTH (Read-only): Displays the active financial year and month (e.g., "2026/01") for which the bonuses are being finalized.
- Helper Text & Routing Logic: Provides critical system instruction: "Click 'Process Bonuses' to update the bonus records for this month. Entries marked 'Pay with Salary' will be added to the employee's salary." * Business Logic Note: This confirms that the engine evaluates the "Pay w/ Salary" checkbox set during the Entry phase. If flagged true, the processed amount is queued into the main Salary Transaction Process. If flagged false, the amount remains isolated and must be extracted via a separate bank file generation.



Module: Masterdata Uploading Process, this interface serves as a bulk data management utility. It allows system administrators to bypass manual, individual record creation by importing large datasets directly into the core database tables. This is primarily used during initial system setup, mass organizational restructuring, or large-scale data migrations.

Core Upload Logic & Behavior This module operates on an "Upsert" (Update or Insert) logic, governed by the following rules outlined in the system helper text:

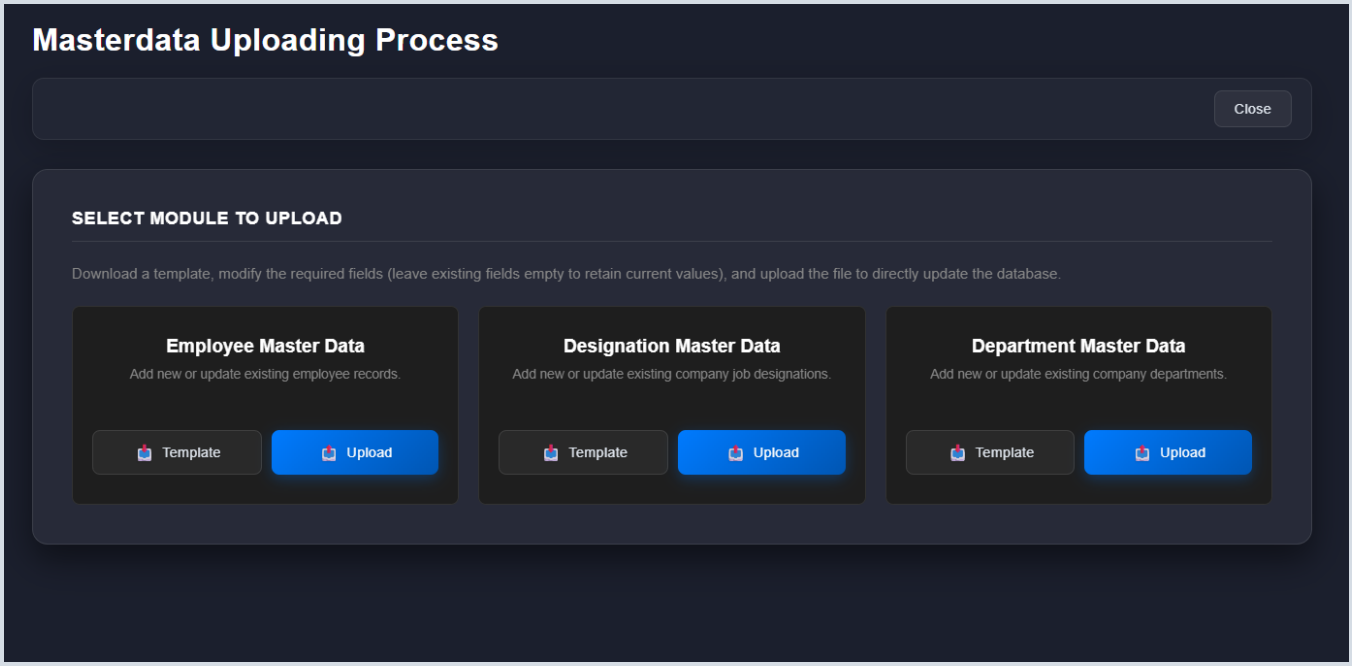
- Insertions: Adding entirely new rows to the uploaded spreadsheet will create new records in the database.
- Updates: Modifying data for existing records (matched via primary keys like Employee Code or Department Code) will overwrite the old database values.
- Partial Updates (Blank Field Logic): If an existing record's field is intentionally left blank in the uploaded template, the backend engine will ignore that cell and retain the current value already stored in the database, preventing accidental data erasure.

Supported Master Data Modules The interface categorizes uploads into specific entity cards, each containing its own dedicated template and upload handlers.

- Employee Master Data: Used to mass-import new hires or perform bulk updates to existing employee profiles (e.g., mass department transfers or widespread salary adjustments).
- Designation Master Data: Used to quickly populate or revise the standardized list of organizational job roles.
- Department Master Data: Used to establish or restructure the company's internal divisions and cost centers.

Module-Specific Action Controls (Per Card) Each module card utilizes a standard pair of buttons to facilitate the upload cycle:

- Template (Button): Generates and downloads a standardized, specifically formatted spreadsheet (e.g., CSV or Excel) containing the exact database column headers required for that specific module. This ensures data is structured correctly before ingestion.
- Upload (Button): Opens a file directory prompt allowing the user to select their completed template. Upon selection, this triggers the backend parsing script, validates the data against system rules, and executes the database upsert commands.



Module: Month End Process, this interface executes the final, irreversible step of the monthly payroll cycle. Once all salaries, bonuses, and statutory files have been processed, tallied, and disbursed, this module is used to "close the books" for the current period. It archives the finalized data, clears the active transaction entry grids, and rolls the system forward to the next financial month.

Action Bar

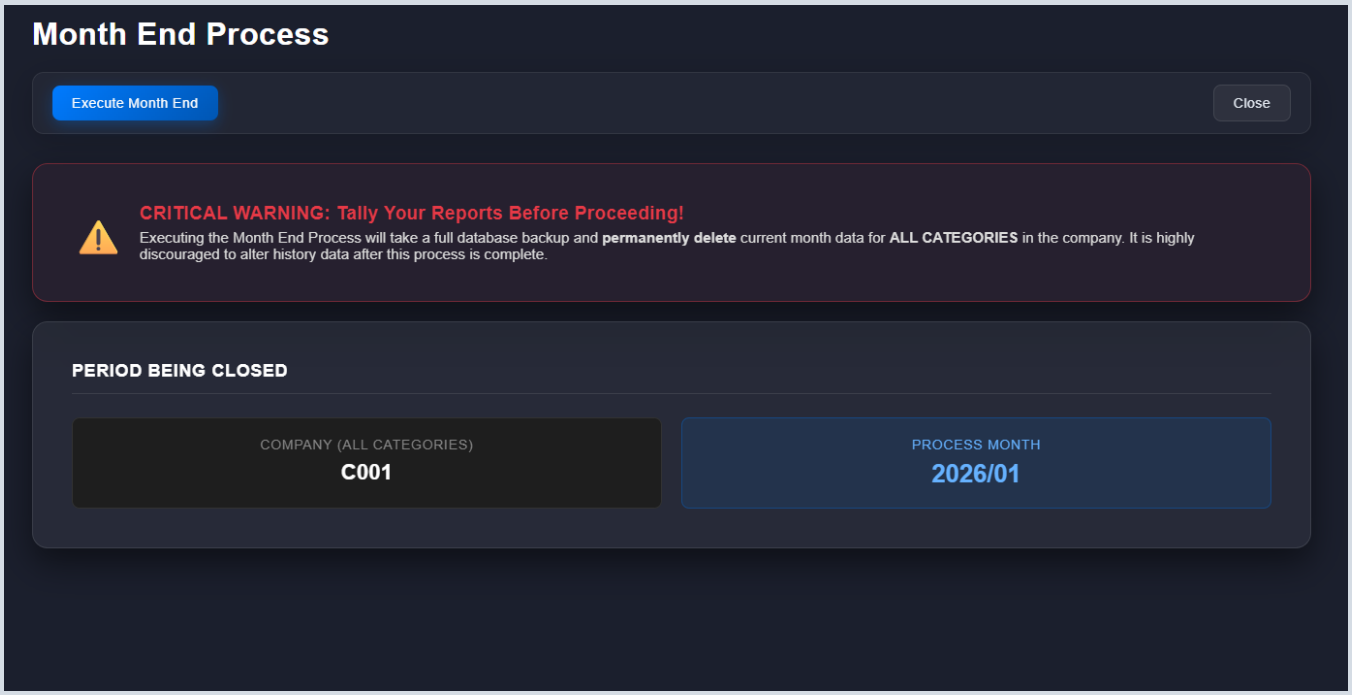
- Execute Month End (Button): The primary, highly restricted trigger command. Clicking this initiates the backend rollover script.
- Close (Button): Exits the module safely without triggering any scripts.

Critical Warning Banner This is a prominent UI safety feature designed to prevent accidental execution.

- System Warning & Behavior: It explicitly instructs the administrator to ensure all management reports are tallied prior to execution. It details the backend actions: the system will trigger an automated database backup, and then it will clear (or "permanently delete" from the active operational tables) the current month's transaction data, moving the finalized figures into historical or Year-to-Date (YTD) storage archives.

Period Being Closed Section This section acts as a final read-only confirmation of the scope of the rollover before the user clicks execute.

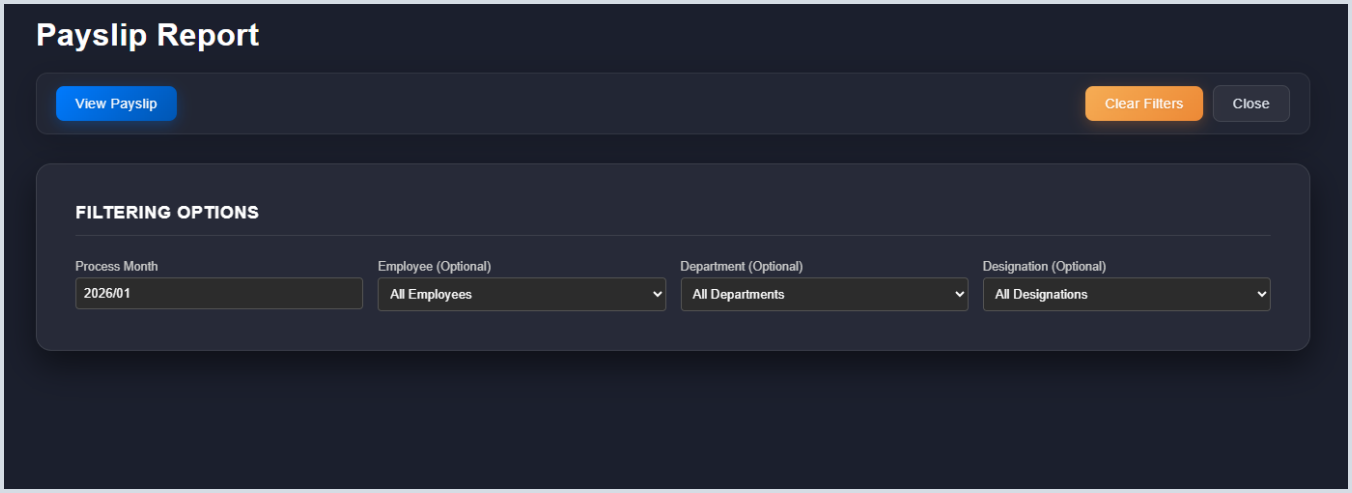
- COMPANY (ALL CATEGORIES): Displays the specific Company Code (e.g., "C001") being closed. Business Logic Note: Unlike the salary calculation modules which can be run in batches by employee category, the Month End process is a global, company-wide execution. It locks the period for all categories simultaneously.
- PROCESS MONTH: Displays the exact active financial year and month (e.g., "2026/01") that is about to be archived and closed out. Once executed, the global system period will automatically increment to the next sequential month (e.g., 2026/02).



Module: Payslip Report, This interface acts as the parameter selection screen for generating employee payslips. It allows administrators or HR personnel to query the finalized payroll data and extract either individual or batch payslips based on specific organizational filters.

Filtering Options Section This section dictates exactly which employees will be included in the generated report. By combining these filters, administrators can run highly targeted reports (e.g., pulling payslips only for "Managers" in the "IT Department").

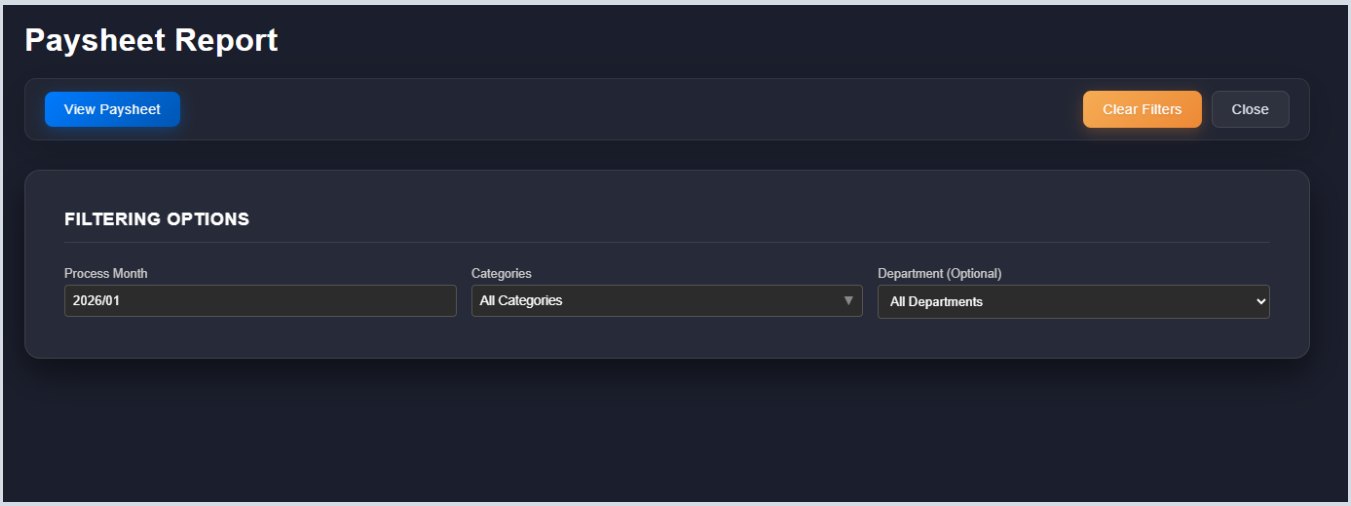
- Process Month (Mandatory): Defines the specific financial year and month (e.g., "2026/01") for which the payslips should be generated.
- Employee (Optional) (Dropdown): Allows the user to select a specific individual employee to generate a single payslip. If left at the default "All Employees", the system will generate payslips for the entire selected cohort.
- Department (Optional) (Dropdown): Allows the system to filter and generate batch payslips grouped by a specific organizational department (e.g., printing all payslips for the "Human Resources" team).
- Designation (Optional) (Dropdown): Allows the system to filter and generate batch payslips based on a specific job role or title across the company.



Module: Paysheet Report, this interface is used to generate the comprehensive payroll register (often referred to as the master paysheet). Unlike individual payslips, this report provides a sprawling, line-by-line grid detailing every single employee's earnings, statutory deductions, loan recoveries, and final net pay for the selected period. It serves as the primary document for internal financial auditing and detailed payroll reconciliation.

Filtering Options Section This section allows administrators to define the scope of the generated paysheet.

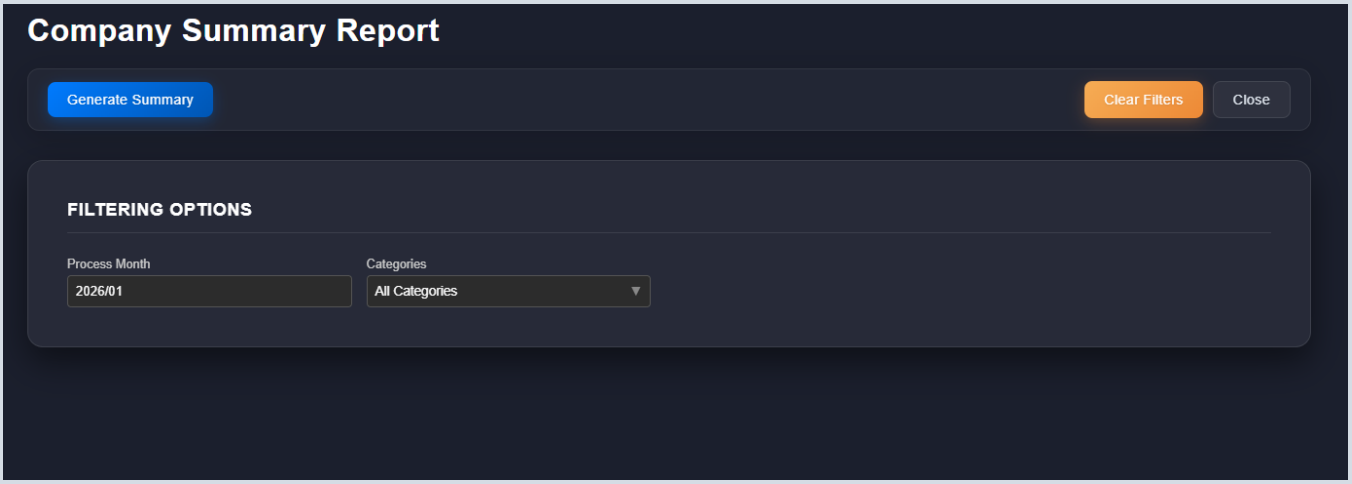
- Process Month (Mandatory): Defines the specific financial year and month (e.g., "2026/01") to be audited.
- Categories (Dropdown): Allows the user to filter the report by specific employee classifications. If set to "All Categories", it generates a master sheet for the entire company.
- Department (Optional) (Dropdown): Allows the system to further narrow the paysheet down to a specific division or cost center, which is highly useful for departmental budget tracking.



Module: Company Summary Report, this interface generates a high-level, aggregated financial overview of the payroll run. Unlike the Paysheet Report, the Company Summary omits individual employee rows. Instead, it calculates and displays macro-level totals (e.g., Total Basic Salary Expense, Total EPF Employer Contributions, Total Net Pay Disbursed). This report is primarily used by the finance department to quickly post consolidated payroll journal entries into the company's main General Ledger (GL) system.

Filtering Options Section This section is intentionally more streamlined than the detailed paysheet filters, focusing on high-level financial groupings.

- Process Month (Mandatory): Defines the specific financial period to be summarized.
- Categories (Dropdown): Allows finance personnel to pull aggregated totals for the entire company ("All Categories") or isolate the macro-level costs associated with a specific employee group (e.g., comparing the total payroll expense of "Management" vs "Contract" staff).



Module: Individual Summary Reports, this interface is a highly customizable reporting tool designed for analytical tracking and variance analysis. It allows administrators to build ad-hoc reports by selecting a specific date range and hand-picking exact salary components (e.g., tracking just "Attendance Allowance" and "Overtime" over a 6-month period for a specific department).

Filtering Options Section This section defines the timeline and the subject scope of the custom report.

- From Month & To Month (Mandatory): Unlike standard single-month reports, these fields define a date range (e.g., from 2026/01 to 2026/06). This is crucial for multi-month trend analysis or quarterly auditing.
- Categories (Dropdown): Filters the dataset to include only specific employee classifications.
- Group By (Rows) (Dropdown): Dictates the primary structure of the generated report. Selecting "Employee" will list individual staff members as rows.
- Employee (Optional) (Dropdown): Allows the user to narrow the multi-month query down to one specific individual's payment history.

Select Columns (Salary Items) Grid This section acts as a dynamic column builder for the final report. Only the items checked here will appear as data columns in the generated output.

- Master Checkbox (Top Left): Toggles all visible salary items on or off simultaneously.
- Item Description: The human-readable name of the salary component (e.g., "Basic Salary", "Stamp Duty").
- Type: A categorized badge (e.g., EARNINGS, DEDUCTIONS) visually indicating how the component affects the employee's gross pay.
- System Code: The backend alphanumeric identifier (e.g., "SI001", "BASIC") associated with the salary item in the master database.
- Row Checkbox: Allows the user to individually select or deselect specific items. In the provided example, only "Attendance Allowance" and "Basic Salary" are checked, meaning the final report will ignore all other deductions or allowances and strictly map out those two financial figures against the chosen filters.

Individual Summary Reports

Generate Report | Clear Items | Close

FILTERING OPTIONS

From Month	To Month	Categories	Group By (Rows)	Employee (Optional)
2026/01	2026/01	All Categories	Employee	All Employees

SELECT COLUMNS (SALARY ITEMS)

☐	Item Description	Type	System Code
☐	Stamp Duty	DEDUCTIONS	STAMP
☐	Welfare Deduction	DEDUCTIONS	SI101
☑	Attendance Allowance	EARNINGS	SI001
☑	Basic Salary	EARNINGS	BASIC
☐	Basic Salary (Pre-Increment)	EARNINGS	BASIC
☐	Fixed Allowance	EARNINGS	SI102

ABC SYSTEMS (PVT) LTD

INDIVIDUAL SUMMARY REPORT - GROUPED BY EMPLOYEE

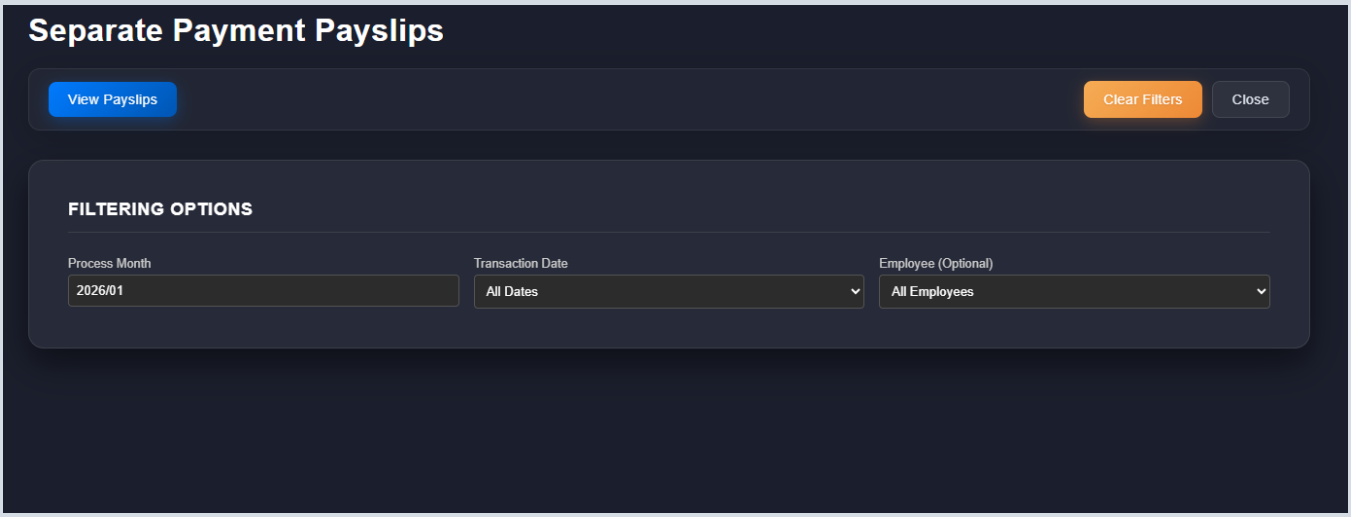
PERIOD: 2026/01 to 2026/01 | CATEGORY: C003

ID	Employee Name	EPF Employee 8%	Basic Salary	Fixed Allowance
EMP004	Ann Smiths	14,300.00	50,000.00	150,000.00
EMP021	Ramesh Ilangasinghe	9,867.90	-	-
EMP022	Malini Wickramasinghe	7,658.00	82,000.00	4,500.00
	TOTALS:	31,825.90	132,000.00	154,500.00

Module: Separate Payment Payslips, this interface is used to generate individual payment advice slips specifically for ad-hoc disbursements (e.g., specific expense reimbursements, travel advances, or severance) that were processed independently of the main payroll. Distributing these separate slips ensures clear communication with employees regarding non-standard deposits.

Filtering Options Section Because separate payments are event-driven rather than strictly monthly, this section includes a specific date filter.

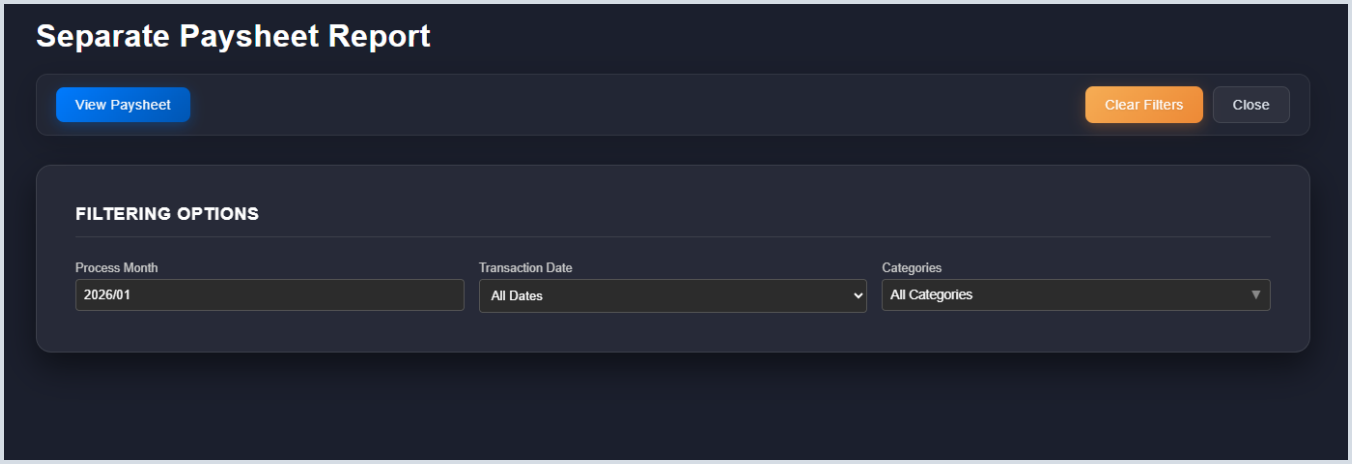
- Process Month (Mandatory): Defines the overarching financial year and month (e.g., "2026/01") in which the ad-hoc payments were processed.
- Transaction Date (Dropdown): A critical filter unique to separate payments. Since a company might run multiple ad-hoc payment batches in a single month (e.g., expenses on the 5th, severances on the 20th), this allows the administrator to isolate and generate slips only for the specific date the separate payment was executed.
- Employee (Optional) (Dropdown): Allows the generation of a separate payslip for one specific individual, or defaults to "All Employees" to generate a batch for everyone included in that specific transaction date's run.



Module: Separate Paysheet Report, similar to the main Paysheet Report, this interface generates the comprehensive, tabular master register, but strictly for ad-hoc "Separate Payments." It provides a consolidated view of all isolated disbursements for auditing and financial reconciliation before the separate bank transfer file is authorized.

Filtering Options Section

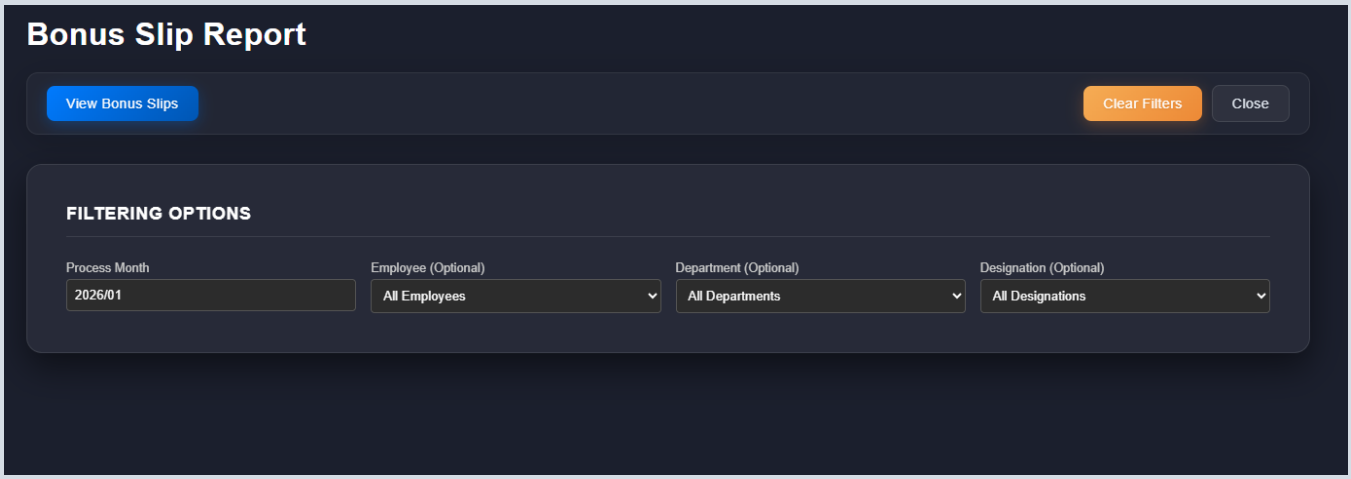
- Process Month (Mandatory): The financial period being audited.
- Transaction Date (Dropdown): Allows the finance team to pull a master sheet for a specific ad-hoc payment run based on the exact day it was processed.
- Categories (Dropdown): Allows administrators to filter the separate payment master sheet by specific employee classifications (e.g., auditing separate travel allowances strictly for the "Executive" category).



Module: Bonus Slip Report, this interface is dedicated to generating payment slips specifically for processed bonuses. Since bonuses often involve distinct tax rules (APIT Bonus tables) and are frequently disbursed in independent bank runs, generating dedicated bonus slips provides clarity to employees regarding their performance rewards and corresponding tax deductions.

Filtering Options Section This section utilizes standard demographic routing to target the exact group receiving the bonus documentation.

- Process Month (Mandatory): Defines the financial period when the bonus was calculated and processed.
- Employee (Optional) (Dropdown): Allows targeted generation of a bonus slip for a single individual.
- Department (Optional) (Dropdown): Allows the system to generate batch bonus slips for a specific division (e.g., printing slips only for the "Sales" department after a quarterly commission run).
- Designation (Optional) (Dropdown): Allows filtering by a specific job role to isolate who receives the printed or PDF bonus documentation.



Module: Loan Detail Report, this interface is a specialized auditing tool dedicated exclusively to tracking employee credit facilities and loan recoveries. It allows human resources to view individual repayment histories and enables the finance department to reconcile aggregate loan deductions for a specific payroll period.

Report Type: Employee Wise (Installment History) This toggle state switches the interface into a historical ledger view. It is designed to pull the lifetime repayment timeline for specific individuals, showing how much principal has been recovered and the remaining outstanding balance.

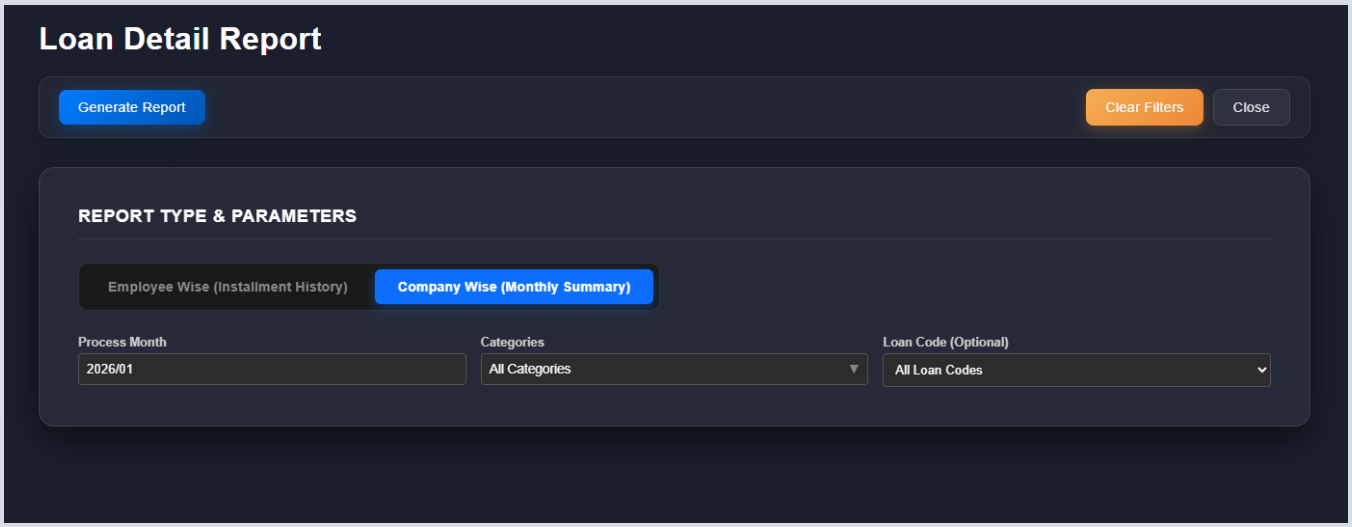
Filtering Options:

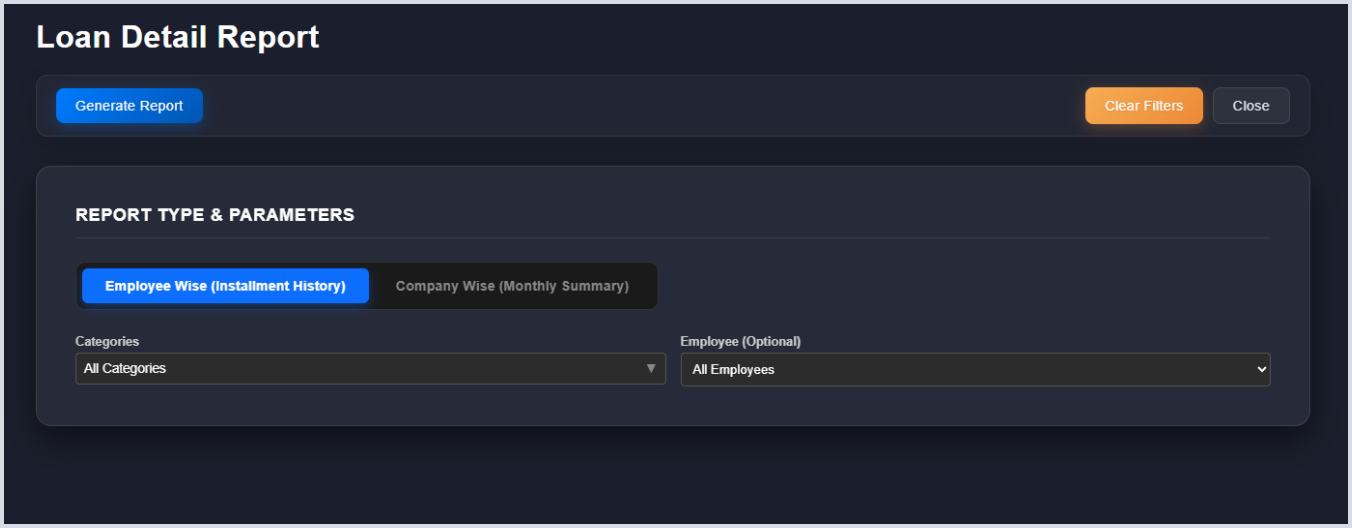
- Categories (Dropdown): Allows the user to filter the historical search by specific employee classifications.
- Employee (Optional) (Dropdown): The primary filter for this view. It allows the administrator to select a specific individual and generate a complete historical statement of their active or past loans.

Report Type: Company Wise (Monthly Summary) This toggle state switches the interface into a macro-level financial view. Instead of historical timelines, it isolates a single payroll run and aggregates the total loan deductions withheld across the company during that specific month.

Filtering Options:

- Process Month (Mandatory): Defines the specific financial year and month (e.g., "2026/01") to be audited.
- Categories (Dropdown): Allows finance personnel to pull aggregated loan totals for the entire company or isolate specific employee groups.
- Loan Code (Optional) (Dropdown): Allows the administrator to filter the monthly summary by a specific type of credit facility defined in the Loan Master File (e.g., generating a report strictly for "Vehicle Loan" deductions processed in January 2026).


Loan Detail Report
Generate Report
Clear Filters
Close
REPORT TYPE & PARAMETERS
Employee Wise (Installment History)
Company Wise (Monthly Summary)
Process Month
2026/01
Categories
All Categories
Loan Code (Optional)
All Loan Codes


Loan Detail Report
Generate Report
Clear Filters
Close
REPORT TYPE & PARAMETERS
Employee Wise (Installment History)
Company Wise (Monthly Summary)
Categories
All Categories
Employee (Optional)
All Employees

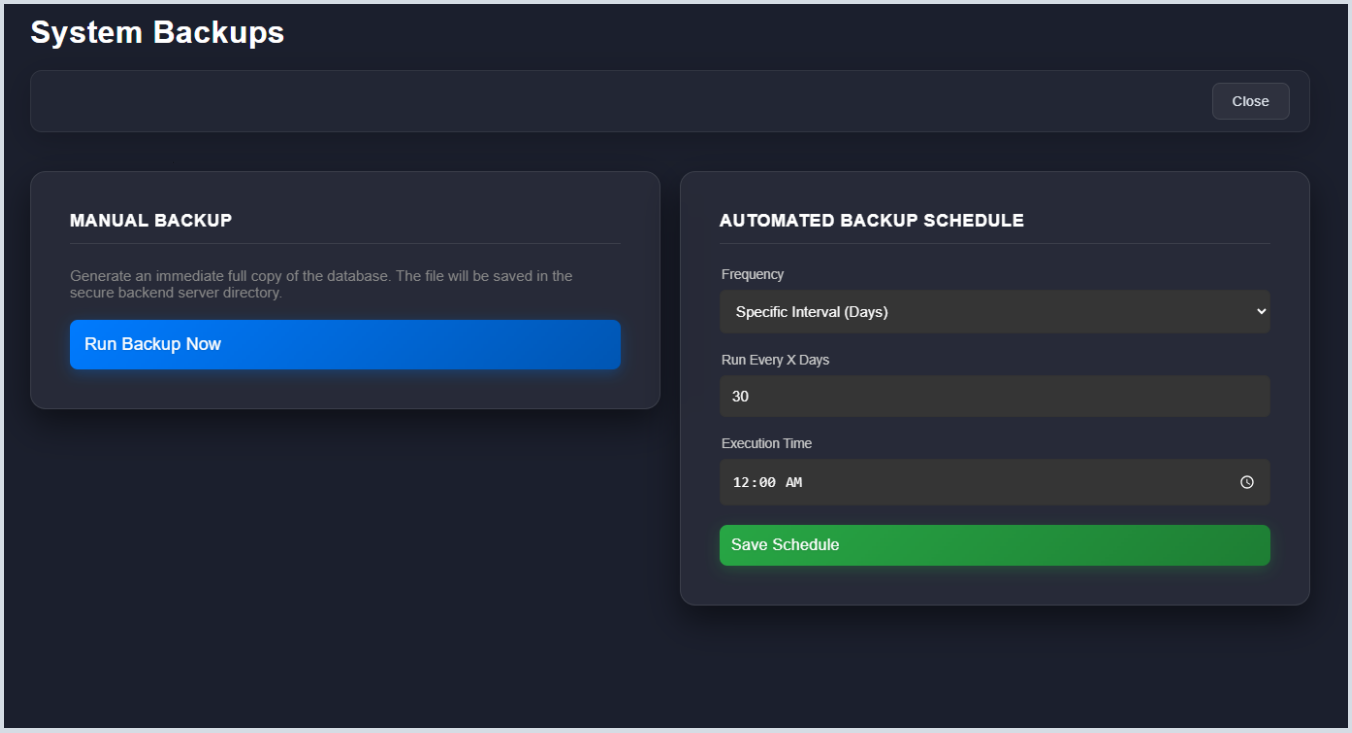
Module: System Backups, this interface manages the disaster recovery and data preservation protocols for the payroll system. It provides administrators with the tools to either instantly generate a snapshot of the database or configure a recurring, automated schedule to ensure data is consistently backed up without manual intervention.

Manual Backup Section This section is used for immediate, ad-hoc data preservation. It is typically utilized right before executing major system changes, such as the "Month End Process" or a mass data upload.

- Run Backup Now (Button): The execution command. Clicking this immediately triggers the backend server to compile a full copy of the current database and save it securely in the designated server directory.

Automated Backup Schedule Section This section configures the system's automated cron jobs, dictating when and how often the server should automatically back up the data.

- Frequency (Dropdown): Defines the primary cadence for the automated backups. The available options include:
    - Every Day: Executes a daily backup.
    - Every Week (Sunday): Executes a weekly backup at the end of the standard week.
    - Specific Interval (Days): Allows for a custom, recurring timeline.
- Run Every X Days (Input Field): This field dynamically appears based on the Frequency selection (specifically when "Specific Interval" is chosen). It allows the administrator to define the exact numeric interval (e.g., entering "30" will trigger the backup every 30 days).
- Execution Time (Time Picker): Defines the exact hour and minute the automated backup script will run. Best practice dictates setting this during off-peak hours (e.g., "12:00 AM") to prevent server load during active payroll processing times.
- Save Schedule (Button): Commits the configured parameters to the server, activating the automated backup routine.


System Backups
Close
MANUAL BACKUP
Generate an immediate full copy of the database. The file will be saved in the secure backend server directory.
Run Backup Now
AUTOMATED BACKUP SCHEDULE
Frequency
Specific Interval (Days)
Run Every X Days
30
Execution Time
12:00 AM
Save Schedule

Future Enhancements (Out of Scope)

While the current system architecture fully addresses the core payroll calculation, transaction management, and statutory reporting requirements, the following features have been identified as valuable enhancements for future development phases. These items remain strictly out of scope for the current release but are planned for the product roadmap:

- **Dedicated Attendance & Leave Management Module:** * Currently, attendance allowances and deductions are handled via bulk uploads or manual transaction entries. A future iteration will introduce a dedicated module capable of direct integration with biometric time-clock devices.
    - This will include automated shift management, automated Overtime (OT) extraction, and a robust leave application system that natively syncs with the payroll calculation engine.
- **Interactive Management Dashboard:**
    - The implementation of a centralized, visually rich analytical dashboard for system administrators and company executives.
    - This dashboard will feature a premium, modern aesthetic (utilizing advanced UI elements like dark themes and glassmorphism) to display high-level payroll metrics, graphical trend analyses of month-over-month expenses, and quick-access widgets for pending approvals or recent backups.
- **Employee Self-Service (ESS) Portal:**
    - A secure, separate web portal allowing individual employees to log in independently. Employees will be able to view and download their historical payslips, check their EPF balances, and view personal loan installment histories without requiring HR intervention.
- **Future-Proofing & API Integration Architecture:**
    - Expanding the system's backend to support RESTful API endpoints, allowing the payroll software to seamlessly communicate with third-party Enterprise Resource Planning (ERP) systems, specialized HR software, or automated accounting ledgers.
    - Continued optimization of the relational database structure and backend logic to ensure the system remains highly scalable and performant as the organization's employee headcount grows significantly.

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